

Associazione Italiana Financial Industry Risk Managers

Asset & Liability Management in Modern Banking

**Theory, Applications,
and Case Studies**

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Asset & Liability Management in Modern Banking

**Theory, Applications,
and Case Studies**

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Asset & Liability Management in Modern Banking: Theory, Applications, and Case Studies

Table of Contents

Part I – Theory and Framework

Chapter 1 – Introduction to Asset & Liability Management	Pag. 8
Chapter 2 – Balance Sheet Dynamics and Interest Rate Sensitivity	Pag. 12
Chapter 3 – Interest Rate Risk Models and Measurement Frameworks	Pag. 17
Chapter 4 – Liquidity Risk and Funding Strategy in ALM	Pag. 23
Chapter 5 – Balance Sheet Optimization and Strategic ALM	Pag. 30
Chapter 6 – Interest Rate Risk in the Banking Book (IRRBB)	Pag. 36
Chapter 7 – Behavioral Modeling and Non-Maturing Deposits (NMDs)	Pag. 42
Chapter 8 – Liquidity Risk Management and Funding Strategy	Pag. 48
Chapter 9 – Funds Transfer Pricing (FTP) and Internal Capital Allocation	Pag. 55
Chapter 10 – Capital Management, ICAAP, and SREP Framework	Pag. 62
Chapter 11 – Liquidity Risk Management and ILAAP	Pag. 69
Chapter 12 – Integration of ICAAP and ILAAP: The Holistic Balance Sheet View	Pag. 75

Part II – Applications and Case Studies

Chapter 13 – Case Study 1: Interest Rate Risk and ALM Optimization under Monetary Tightening **Pag. 83**

Chapter 14 – Case Study 2: Liquidity Stress and Funding Strategy under Market Disruption **Pag. 88**

Chapter 15 – Case Study 3: Managing Non-Maturing Deposits (NMDs) and Behavioral Optionality in a Low-Rate Environment **Pag. 93**

Chapter 16 – Case Study 4: Duration Gap Management and the Trade-Off Between ΔEVE and ΔNII Sensitivities **Pag. 98**

Chapter 17 – Case Study 5: Liquidity Buffer Optimization and the LCR/NSFR Trade-Off **Pag. 103**

Chapter 18 – Case Study 6: Hedging Fixed-Rate Assets under the IFRS 9 Macro Fair Value Hedge Framework **Pag. 109**

Chapter 19 – Case Study 7: Behavioral Modelling of Non-Maturing Deposits (NMDs) and Interest Rate Sensitivity **Pag. 115**

Conclusion – The Evolution of ALM: From Static Measurement to Strategic Intelligence **Pag. 120**

ABOUT THE AUTHOR **Pag. 123**

Executive Summary

Asset & Liability Management in Modern Banking: Theory, Applications, and Case Studies explores the profound transformation of balance sheet management in an era of monetary instability, regulatory tightening, and digital acceleration.

Bridging academic rigor and practical insight, the book examines how Asset & Liability Management (ALM) has evolved from a traditional risk-measurement discipline into a strategic intelligence function at the heart of modern financial institutions.

The first part of the volume develops a comprehensive theoretical and methodological framework:

- Interest rate risk, convexity, and basis exposure;
- Behavioral modelling of non-maturing deposits and prepayment risk;
- Integration of ALM with liquidity and capital management;
- The role of hedge accounting (IFRS 9) and macro fair value hedging in stabilizing earnings and regulatory capital.

The second part applies these concepts through empirical case studies and scenario simulations, drawing on recent market events — from the Global Financial Crisis to the 2023 U.S. regional bank turmoil.

Each case illustrates how ALM functions can identify vulnerabilities, design effective hedging structures, and optimize balance sheet resilience across rate cycles.

Throughout, the book emphasizes that effective ALM is not about forecasting, but about preparing — through data-driven, scenario-based, and governance-integrated frameworks that align profitability, liquidity, and stability.

Combining analytical depth with real-world relevance, *Asset & Liability Management in Modern Banking* serves as both a reference manual for practitioners and a didactic foundation for scholars and policymakers seeking to understand how banks can transform balance sheet risk into strategic opportunity in the next generation of finance.

PART I

Theory and Framework

Chapter 1 – Introduction to Asset & Liability Management

1.1 Rationale and Definition of ALM

In the contemporary banking environment, **Asset & Liability Management (ALM)** represents a core strategic discipline that bridges the gap between risk management, balance sheet optimization, and long-term profitability. ALM entails the integrated management of a financial institution's assets (e.g. loans, securities) and liabilities (e.g. deposits, borrowings) so as to maintain stability in earnings, capital, and liquidity under variable market conditions.

As defined in the literature, ALM is “the process of formulating, implementing, monitoring, and revising strategies relating to assets and liabilities to achieve an institution's financial objectives, given its risk tolerances and constraints” (Romanyuk, *Asset and Liability Management: An Overview*, 2010). This holistic view contrasts with fragmented “siloed” approaches that treat interest rate risk, liquidity risk, and funding risk separately.

The keyword in ALM is **matching**—that is, aligning the timing, cash flows, and interest rate sensitivity of assets and liabilities. Poor matching exposes the bank to mismatch losses when interest rates shift or funding stress emerges.

In its broadest sense, ALM lies at the intersection of **strategy, markets, and regulation**. While Treasury and market risk functions may address short-term trading risks, ALM's purview is structural and long-term: steering the bank's core balance sheet through evolving environments.

1.2 Objectives of ALM

The primary objectives of an ALM function within a bank can be summarized as follows:

1. Interest Rate Risk Mitigation

- Control the sensitivity of net interest income (NII) and economic value of equity (EVE) to changes in the interest rate environment.
- Prevent adverse effects from parallel or non-parallel rate shifts, curve twists, and volatility changes.

2. Liquidity & Funding Reliability

- Ensure that the institution can meet its cash obligations under normal and stressed conditions (e.g. deposit withdrawals, maturities).
- Safeguard liquidity buffers and stable funding sources (e.g. retail deposits, secured funding, wholesale markets).

3. Capital Preservation & Return Optimization

- Safeguard the bank's capital base by avoiding undue mark-to-market losses and by adhering to regulatory constraints (e.g. Basel, ICAAP, IRRBB).
- Seek to optimize spread income, risk premia, and ALM contributions to the return on regulatory capital.

4. Regulatory Compliance & Governance

- Comply with supervisory requirements (e.g. IRRBB guidelines, liquidity mandates, stress testing).
- Provide transparent reporting and risk oversight to the ALCO, Board, and regulatory bodies (for example, as recommended in the PRA's "Asset and Liability Management: Suggestions for Greater Effectiveness").

5. Strategic Steering & Scenario Readiness

- Use scenario analysis and stress testing to anticipate adverse shocks (e.g. rate regime shifts, funding liquidity stress).
- Adjust balance sheet posture proactively rather than reactively.

These objectives must be achieved under realistic constraints: capital adequacy, regulatory limits, internal risk appetite, and funding availability.

1.3 The ALM Function and Organizational Placement

The effectiveness of ALM hinges on both technical models and sound governance. Key organizational elements include:

- **ALCO / Treasury Committee:** A senior forum (often composed of CEO, CFO, CRO) that defines ALM strategy, sets limits, and reviews performance. The PRA supervisory guidance suggests that the ALCO should be forward-looking and empowered to deliberate both historic results and scenario-driven decisions. [bankofengland.co.uk](https://www.bankofengland.co.uk)
- **ALM Office / Desk:** Executes the day-to-day modeling, scenario simulations, positioning, and reporting.
- **Risk & Finance Interface:** Tight integration with risk (market, credit, liquidity) teams, as well as finance/accounting (especially for hedge accounting and valuation).
- **Governance & Escalation:** Clear lines of authority, documented minutes, exception reporting, and alignment with ICAAP / ILAAP.

In many banks, ALM is embedded in a broader **Total Bank Management (TBM)** function, which encompasses not only interest rate and liquidity risk but also strategic decisions on credit, capital allocation, and business lines.

1.4 Key Concepts and Metrics

To pave the path for subsequent chapters, here are foundational concepts and common metrics:

- **Repricing Gap / Maturity Gap:** The difference between assets and liabilities that will reprice or mature within a given horizon.
- **Duration Gap:** A more sophisticated measure that weights cash flows by time and discounting, estimating interest rate sensitivity of equity (e.g. duration (assets) – (liabilities × assets/liabilities ratio)).
- **ΔEVE (Change in Economic Value of Equity):** The projected change in the net present value of the bank’s assets minus liabilities under a rate shock.
- **ΔNII (Change in Net Interest Income):** The expected variation in interest income minus interest cost over a horizon (often 1 year) under different rate scenarios.
- **Liquidity Gap & Cash Flow Mismatch:** Assessing when inflows and outflows may create funding stress.
- **Behavioral Modeling & Non-Maturing Balances:** Recognizing that not all deposits behave according to contractual maturities, particularly demand or savings accounts.

Later chapters will explore **convexity risk**, **vega sensitivity**, **hedge accounting**, and **optimization frontiers**, building upon these basic building blocks.

1.5 Structure of the Book and Roadmap

This book is organized into two complementary parts:

- **Part I – Theory & Framework:** establishes rigorous conceptual foundations, definitions, risk measurement models, and governance structures.
- **Part II – Applications & Case Studies:** brings theory to life via real banking episodes, simulations, and strategic decisions.

Each chapter concludes with “**Managerial Insights**” boxes that distill the practical implications of theoretical results.

In the next chapter (Chapter 2), we will dive deeper into **balance sheet dynamics and interest rate sensitivity**, exploring how cash flows, repricing, and durations interact in a real banking book.

Chapter 2 – Balance Sheet Dynamics and Interest Rate Sensitivity

2.1 The Structure of the Bank Balance Sheet

The banking balance sheet reflects a fundamental economic transformation: the conversion of **short-term liabilities** (such as customer deposits or wholesale funding) into **long-term earning assets** (such as loans and securities). This maturity transformation, while essential to intermediation, creates exposure to changes in interest rates and liquidity conditions.

A simplified structure may be represented as follows:

Assets		Liabilities & Equity	
Loans to customers (short- and long-term)	60%	Customer deposits (demand & term)	70%
Investment portfolio (sovereign, corporate, ABS)	20%	Wholesale funding (repos, bonds)	15%
Interbank placements	10%	Central bank operations	5%
Cash and reserves	5%	Equity capital	10%

This schematic balance sheet underscores the structural asymmetry between **asset maturity** (longer, fixed-rate) and **liability maturity** (shorter, floating-rate). It is precisely this asymmetry that ALM seeks to manage.

As the Bank for International Settlements (BIS, *Sound Practices for Managing Interest Rate Risk in the Banking Book*, 2016) notes, “banks’ earnings and economic value are vulnerable to changes in interest rates when the maturity or repricing structure of their assets, liabilities, and off-balance-sheet positions are mismatched.”

2.2 Repricing and Duration Gaps: Core Analytical Tools

(a) The Repricing Gap

The **repricing gap** measures the difference between the volume of assets and liabilities that reprice within a specified time bucket. It is typically used to estimate *short-term* exposure to interest rate changes.

$$\text{Repricing Gap}(t) = \text{RSA}(t) - \text{RSL}(t)$$

Where:

- **RSA** = Rate Sensitive Assets
- **RSL** = Rate Sensitive Liabilities

If the repricing gap is positive, assets reprice faster than liabilities — the bank benefits from rising rates but suffers when rates fall. Conversely, a negative gap implies vulnerability to rising rates.

Although straightforward, this method fails to account for timing and magnitude differences in cash flows — hence the need for a more refined approach.

(b) The Duration Gap

The **duration gap** captures the *weighted average maturity* of assets and liabilities, providing a measure of how the bank's net worth responds to interest rate shocks:

$$DG = D_A - \frac{L}{A} D_L$$

where:

- D_A = duration of assets
- D_L = duration of liabilities
- L/A = leverage ratio (liabilities over assets)

A positive duration gap indicates that the economic value of equity (EVE) will fall when interest rates rise (since assets are longer). A negative gap indicates the opposite.

Empirically, a 1% (100 bps) parallel rise in interest rates leads to an approximate change in EVE given by:

$$\Delta EVE = -DG \times A \times \Delta i / (1 + i)$$

This measure is fundamental to the **IRRBB (Interest Rate Risk in the Banking Book)** supervisory framework defined by the Basel Committee (BCBS #368, 2016; BCBS #496, 2021).

2.3 Economic Value vs. Earnings Perspective

There are two complementary perspectives to assess interest rate risk:

Perspective	Horizon	Measure	Objective
Earnings Perspective (ΔNII)	Short-term (1–2 years)	Change in Net Interest Income	Assess impact on profitability
Economic Value Perspective (ΔEVE)	Long-term (entire cash flow life)	Change in Economic Value of Equity	Assess impact on capital/equity value

While the ΔNII perspective focuses on *income volatility*, the ΔEVE perspective captures *valuation risk*. Together, they provide a holistic picture of how sensitive the bank is to changes in the yield curve.

For example, under the Basel standardized 200 bps shock, a ΔEVE exceeding 15% of Tier 1 capital is considered an outlier and subject to supervisory scrutiny (EBA/GL/2018/02).

2.4 Non-Linearity and Convexity in the Banking Book

Duration assumes a linear relationship between rates and value changes. However, this is only a first-order approximation. In reality, bond and loan values exhibit **convexity**, meaning that the response of EVE to large rate movements is non-linear.

Convexity becomes critical when:

- Interest rate shocks are large (≥ 200 bps);
- Option-like features exist (caps/floors in loans, prepayment options, callable funding);
- Behavioral elements (non-maturing deposits) introduce asymmetry.

Banks with material option risk must complement duration measures with **effective duration** and **convexity adjustments** (see: Doff, *Risk Management for Financial Institutions*, Wiley, 2022).

2.5 Behavioral Modeling: Non-Maturing Deposits (NMDs)

One of the most complex elements in ALM is modeling **non-maturing deposits (NMDs)** — retail or corporate balances with no contractual maturity, such as current accounts or savings accounts.

While these deposits can, in theory, be withdrawn on demand, empirical studies show that only a portion behaves as “core,” stable funding. The behavioral duration of NMDs often ranges from **1 to 5 years**, depending on rate sensitivity and customer base (ECB, *Report on IRRBB Modelling Practices*, 2019).

Key modeling assumptions include:

- **Elasticity to interest rates** (how much deposit volumes change with rate moves);
- **Decay rate** (speed of outflow);
- **Pass-through rate** (extent to which deposit rates adjust to market rates);
- **Segmentation** by customer type (retail vs. corporate).

Incorrect modeling of NMDs was among the key weaknesses identified during the **U.S. regional bank crisis (2023)**, where institutions underestimated behavioral repricing and outflow risks.

2.6 From Measurement to Strategic Positioning

ALM is not only a measurement exercise — it is a **strategic management process**. The **duration gap** and **repricing gap** are diagnostic tools; the real decision lies in how management acts upon them.

Possible strategic stances include:

- **Defensive (short duration)**: limit exposure when rates are expected to rise.
- **Neutral (duration-matched)**: maintain balanced sensitivity.
- **Proactive (long duration)**: extend duration when rate cuts are expected.

In practice, banks combine *hedge accounting*, *derivative overlays* (e.g., IRS, swaptions), and *portfolio reallocation* (e.g., from FVOCI to HTC) to implement such strategies. The trade-offs between capital impact, liquidity, and earnings stability are central to ALCO decision-making.

2.7 Managerial Insight: ALM as Strategic Control

A well-calibrated ALM framework transforms risk measures into actionable intelligence. The interaction between ΔEVE , ΔNII , and regulatory capital forms the bank's *strategic frontier* — a trade-off between resilience and profitability.

Understanding this frontier enables the ALCO to decide whether to absorb, transfer, or hedge structural risks.

2.8 References

- Basel Committee on Banking Supervision (2016, 2021), *Interest Rate Risk in the Banking Book (IRRBB)*, BCBS #368 and #496.
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Chapter 3 – Interest Rate Risk Models and Measurement Frameworks

3.1 Introduction

Interest rate risk (IRR) in the banking book (IRRBB) arises when changes in market interest rates affect a bank's earnings or economic value.

The management of IRRBB is central to Asset & Liability Management (ALM) since it influences both short-term profitability and long-term solvency.

The **Basel Committee on Banking Supervision (BCBS 368, 2016; BCBS 496, 2021)** defines IRRBB as “the current or prospective risk to both the bank's capital and earnings arising from adverse movements in interest rates that affect the banking book positions.”

Measurement frameworks must, therefore, capture both:

- **Earnings-based risk**, usually quantified as the change in *Net Interest Income* (ΔNII); and
- **Value-based risk**, captured by the change in *Economic Value of Equity* (ΔEVE).

These two dimensions jointly represent the bank's sensitivity profile to yield curve shocks.

3.2 Conceptual Framework

The objective of IRR measurement is to assess how interest rate shocks — parallel or non-parallel — affect:

- **Earnings** (through repricing of assets/liabilities and behavioral factors), and
- **Economic Value** (through discounted present value of cash flows).

Dual Perspective of Interest Rate Risk

Dimension	Focus	Horizon	Primary Metric	Purpose
Earnings Perspective	Profitability	1–2 years	ΔNII	Income stability
Economic Value Perspective	Capital adequacy	Long-term	ΔEVE	Equity value resilience

An integrated ALM framework models both views jointly, allowing the ALCO to optimize between short-term performance and long-term capital stability.

3.3 Economic Value of Equity (ΔEVE)

The **Economic Value of Equity** represents the net present value (NPV) of all future cash flows expected from on- and off-balance-sheet positions:

$$EVE = PV(\text{Assets}) - PV(\text{Liabilities})$$

A change in interest rates alters these present values. The ΔEVE is computed as:

$$\Delta EVE = EVE_{shock} - EVE_{base}$$

where EVE_{shock} represents the valuation under the stressed yield curve scenario.

Supervisory EVE Scenarios (Basel/EBA)

Under the standardized Basel IRRBB framework, banks are required to assess ΔEVE under six prescribed yield curve shocks:

Scenario	Description	Type
+200 bps	Parallel upward shift	Steepening
-200 bps	Parallel downward shift	Flattening
+Short / -Long	Short rates up, long rates down	Short steepener
-Short / +Long	Short rates down, long rates up	Short flattener
+Short only	Short-term increase	Twist
-Short only	Short-term decrease	Twist

The **Supervisory Outlier Test (SOT)** flags banks with $\Delta EVE > 15\%$ of Tier 1 capital as materially exposed (EBA/GL/2018/02).

3.4 Net Interest Income (ΔNII)

The ΔNII measure assesses the short-term earnings sensitivity to interest rate movements, typically over a 12-month horizon.

Unlike ΔEVE , it reflects **cash flow mismatches** rather than valuation effects.

$$\Delta NII = NII_{shock} - NII_{base}$$

ΔNII captures how much the bank's net interest margin (NIM) would change given a rate shock, considering:

- Timing of repricing for variable-rate instruments;
- Behavioral assumptions (e.g., prepayments, deposit betas);
- Optionality embedded in assets and liabilities.

ΔNII is crucial for understanding *profit resilience* — particularly relevant for banks heavily reliant on maturity transformation.

3.5 Duration and PV01 Approaches

The **Duration-based approach** is a first-order approximation of ΔEVE :

$$\frac{\Delta EVE}{EVE} \approx -DG \times \Delta i$$

where **DG (Duration Gap)** is defined as in Chapter 2.

However, for practical computation, many institutions employ the **PV01** (*Price Value of a Basis Point*) — the change in the present value of a position for a 1 bp change in yield:

$$PV01 = \frac{d(PV)}{d(i)} \times 0.0001$$

For portfolio-level aggregation:

$$\Delta EVE = \sum_i PV01_i \times \Delta i$$

The PV01 approach is used in both internal ALM models and regulatory stress tests as a more granular, instrument-level sensitivity measure.

3.6 Simulation and Dynamic NII Modelling

While static gap and duration models provide useful snapshots, **dynamic simulation models** offer a forward-looking perspective.

These models simulate future NII under various yield curve scenarios, behavioral responses, and balance sheet strategies.

A typical NII simulation framework includes:

1. **Baseline projection** (assuming stable balance sheet and rates);
2. **Interest rate scenarios** (parallel, non-parallel, stochastic);
3. **Behavioral adjustments** (loan prepayments, deposit outflows, pass-through rates);

4. **Strategic actions** (hedging, repricing, reinvestment strategies).

Dynamic simulations can be **deterministic** (based on fixed rate paths) or **stochastic** (Monte Carlo, Hull-White or CIR models), integrating volatility and convexity effects.

3.7 Behavioral Modelling of Optionality

Many banking book positions contain **embedded options**, such as:

- Fixed-rate loans with prepayment rights;
- Callable term deposits or wholesale funding;
- Caps and floors in variable-rate products.

Ignoring these features can materially distort both ΔEVE and ΔNII estimates. Hence, **effective duration** and **option-adjusted spread (OAS)** techniques are employed to adjust for embedded convexity.

Advanced approaches model these using **option pricing models** (e.g., Black, 1976; Hull, 2017) or **behavioral econometrics** (ECB, 2019).

The goal is to capture asymmetric sensitivity: for example, prepayment risk increases when rates fall, reducing effective duration.

3.8 Stress Testing and Scenario Analysis

Regulatory frameworks require banks to conduct **regular IRR stress tests**, exploring extreme yet plausible rate movements.

A comprehensive stress-testing program should include:

- **Parallel and non-parallel shifts** (steepeners, flatteners, twists);
- **Basis risk scenarios** (e.g., widening between Euribor and OIS);
- **Behavioral stresses** (deposit outflows, accelerated prepayments);
- **Convexity shocks** (non-linear sensitivity to large rate changes).

Scenario design should be consistent with the bank's **Risk Appetite Framework (RAF)** and **Internal Capital Adequacy Assessment Process (ICAAP)**.

For instance, under a +200 bps parallel shock, a bank with a DG = 0.5 years and EVE of €10bn would experience:

$$\Delta EVE = -0.5 \times 10,000 \text{ million} \times 0.02 = -100 \text{ million EUR}$$

This ΔEVE would represent a 1% reduction in EVE, well within regulatory thresholds — but the same shock could imply a ΔNII deterioration depending on asset-liability composition.

3.9 Integration with Risk Governance and ALCO Decision-Making

Quantitative results from IRR models must be interpreted within the governance structure. ALM Committees (ALCO) use these outputs to define **strategic duration targets, hedging strategies**, and **funding mix** adjustments.

Integration steps include:

- Mapping $\Delta\text{EVE}/\Delta\text{NII}$ outputs to risk appetite thresholds;
- Evaluating hedge effectiveness under IFRS 9;
- Linking ALM metrics to ICAAP and ILAAP;
- Setting transfer pricing mechanisms reflecting duration and liquidity costs.

Effective governance ensures that IRR measurement is not a compliance exercise but a strategic lever influencing **pricing, funding, and investment**.

3.10 Emerging Topics: Convexity Risk and Non-Linear Exposures

Recent crises (e.g., the 2023 U.S. regional bank failures) have underscored the limits of linear IRR models.

Convexity risk, defined as the sensitivity of duration itself to rate changes, becomes material when yield volatility increases.

A practical market proxy for convexity exposure is **YIV-Beta** — the sensitivity of a bank's stock returns to changes in *yield implied volatility* (Cremers, Fleckenstein & Gandhi, 2021). Banks with high YIV-Beta exhibit stronger vulnerability to rate uncertainty, as in the case of **Silicon Valley Bank (SVB)** prior to its collapse.

Integrating convexity analysis into ALM frameworks — through stress testing of yield volatility and option-adjusted metrics — represents the frontier of interest rate risk management.

3.11 References

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Chapter 4 – Liquidity Risk and Funding Strategy in ALM

4.1 Introduction

Liquidity risk is one of the most fundamental components of the Asset & Liability Management (ALM) framework.

While interest rate risk and credit risk shape a bank's *profitability* and *valuation*, liquidity risk determines its *survival*.

As highlighted by the **Basel III framework** (BCBS 238, 2013), the 2007–2009 Global Financial Crisis revealed how the erosion of market liquidity and the over-reliance on short-term wholesale funding could trigger systemic crises even among solvent institutions.

Liquidity management in ALM involves balancing **liquid asset holdings** with **stable funding structures**, ensuring that the bank can meet its obligations as they come due — both in normal times and under stress.

This chapter explores the conceptual, quantitative, and strategic dimensions of liquidity risk management within modern banking, integrating regulatory and internal perspectives.

4.2 Types of Liquidity Risk

Liquidity risk manifests in multiple, interrelated forms:

Type	Definition	Primary Drivers
Funding Liquidity Risk	The inability to meet obligations when due without incurring unacceptable losses.	Maturity mismatch, deposit volatility, market confidence.
Market Liquidity Risk	The inability to liquidate assets without significant price discounts.	Market depth, bid-ask spreads, asset concentration.
Contingent Liquidity Risk	Unexpected cash outflows or loss of funding capacity due to off-balance-sheet events.	Credit line drawdowns, margin calls, reputational shocks.

The ALM function typically focuses on **funding liquidity risk**, but must monitor market liquidity dynamics to ensure that balance sheet assets designated as “liquid” can indeed be monetized during stress periods.

4.3 Liquidity Metrics: LCR and NSFR

The Basel III liquidity standards established two complementary regulatory metrics to ensure both short-term and structural resilience:

(a) Liquidity Coverage Ratio (LCR)

Ensures that banks maintain a sufficient stock of **High-Quality Liquid Assets (HQLA)** to cover **net cash outflows over a 30-day stress scenario**:

$$LCR = \frac{HQLA}{Net\ Cash\ Outflows_{30d}} \geq 100\%$$

HQLA Composition:

- Level 1 assets: cash, central bank reserves, sovereign bonds (0% haircut);
- Level 2A/2B: covered bonds, corporate debt, equities (subject to caps and haircuts).

The LCR enforces *short-term liquidity discipline*, limiting excessive maturity transformation.

(b) Net Stable Funding Ratio (NSFR)

Promotes *medium-to-long-term structural stability* by requiring banks to maintain sufficient **Available Stable Funding (ASF)** relative to **Required Stable Funding (RSF)**:

$$NSFR = \frac{ASF}{RSF} \geq 100\%$$

ASF Factors: regulatory weights based on funding maturity and stability (e.g., retail deposits receive higher ASF factors than wholesale).

RSF Factors: weights reflecting the liquidity characteristics and residual maturity of assets.

The NSFR complements the LCR by discouraging structural reliance on short-term funding to finance illiquid assets.

4.4 Internal Liquidity Metrics and ALM Dashboards

Beyond regulatory ratios, banks employ a range of **managerial liquidity indicators** to capture nuances not covered by LCR/NSFR. Common examples include:

- **Cumulative Maturity Gaps:** projected net liquidity positions by time bucket;

- **Survival Horizon:** the number of days the bank can meet obligations under stress without external funding;
- **Liquidity Value-at-Risk (LVaR):** probabilistic assessment of extreme liquidity shortfalls;
- **Collateral Utilization Ratio:** share of pledged vs. unencumbered assets;
- **Funding Concentration Indices:** Herfindahl-type measures for counterparty and source diversification.

These metrics feed into the **Liquidity Risk Dashboard**, a core ALM monitoring tool, linking real-time liquidity positions with stress test outputs and funding plan updates.

4.5 Behavioral Modelling of Deposits and Funding Stability

Accurate liquidity projections depend heavily on **behavioral modelling** — particularly for non-maturing deposits (NMDs) and retail funding.

Unlike contractual maturities, behavioral profiles reflect depositor stability, sensitivity to interest rate changes, and stress elasticity.

Key parameters include:

- **Core vs. volatile deposits:** stable vs. rate-sensitive balances;
- **Deposit beta:** elasticity of deposit rates to changes in market rates;
- **Run-off rates:** modeled percentage of withdrawals under stress scenarios.

Empirical studies (e.g., ECB, 2019; EBA, 2020) show that stable retail deposits exhibit average maturities of 3–5 years in behavioral terms — substantially longer than contractual horizons — which supports their treatment as “stable funding” within NSFR.

Behavioral assumptions are validated through *back-testing* and *stress recalibration* during periods of market volatility.

4.6 Liquidity Stress Testing and Contingency Planning

A robust liquidity management framework integrates **periodic stress testing** to assess the institution’s ability to withstand idiosyncratic and systemic shocks.

Core stress test scenarios include:

1. **Institution-specific stress:** loss of wholesale funding, credit rating downgrade, or reputational damage.
2. **Market-wide stress:** systemic liquidity freeze, sudden interest rate shifts, collateral valuation shocks.

3. **Combined stress:** simultaneous idiosyncratic and market triggers.

The results feed into the **Contingency Funding Plan (CFP)**, detailing escalation procedures, funding sources, and governance actions (Board/ALCO triggers).

The CFP ensures readiness for liquidity crises by identifying *early warning indicators* (e.g., LCR < 110%, widening CDS spreads, deposit outflows) and pre-approved liquidity lines.

4.7 Strategic Funding Management within ALM

Liquidity management extends beyond compliance — it is a strategic driver of bank profitability and resilience.

(a) Funding Mix Optimization

The optimal funding mix minimizes cost while ensuring stability:

Funding Source	Advantages	Limitations
Retail deposits	Stable, low cost	Limited scalability
Wholesale funding	Rapid access, flexible	Volatile, confidence-sensitive
Covered bonds	Term funding, lower spreads	Requires eligible collateral
Securitizations	Risk transfer	Complexity, market dependence
Central bank facilities	Liquidity backstop	Stigma, policy dependency

A balanced funding structure aligns maturity profiles and limits reliance on short-term wholesale markets.

(b) Transfer Pricing Mechanisms (FTP)

Funds Transfer Pricing translates liquidity costs and benefits into product-level profitability. Modern FTP frameworks internalize:

- Cost of stable funding (NSFR-based premiums);
- Cost of liquidity buffers (LCR requirements);
- Term liquidity spreads (curve-dependent);
- Behavioral asymmetries (deposit stickiness vs. loan prepayment).

FTP alignment with ALM risk signals ensures coherence between *product pricing*, *liquidity risk*, and *strategic funding plans*.

4.8 Integration of Liquidity and Interest Rate Risk

Liquidity and interest rate risk are inherently interlinked.

An increase in market rates, for example, may reduce deposit stickiness (raising run-off risk), while liquidity stress can lead to forced asset sales that crystallize rate-driven valuation losses. The integration of both dimensions within the ALM governance cycle is thus essential.

Analytical Linkages:

- **Duration–Liquidity Trade-Off:** longer-term funding stabilizes NSFR but increases duration risk;
- **Liquidity–Spread Interaction:** tightening liquidity elevates funding spreads and depresses NII;
- **Asset Liquidity Premiums:** low-liquidity assets require higher returns, affecting portfolio optimization.

Modern ALM systems increasingly employ **Integrated Balance Sheet Simulators (IBS)**, which project EVE, NII, LCR, and NSFR jointly under dynamic scenarios — supporting holistic decision-making.

4.9 Case Illustration: Managing Liquidity in a Rate-Hike Cycle

Consider a mid-sized European bank with:

- LCR = 135%; NSFR = 110%;
- Behavioral core deposits = 70% of retail base;
- Funding gap at 6–12 months bucket = €1.2bn.

Facing a rapid tightening cycle, ALCO assesses two strategic responses:

Option A	Option B
Issue €2bn covered bonds at 5-year maturity	Reprice retail deposits by +40 bps to retain funding base
Improves NSFR to 120%, but compresses NIM by 10 bps	Preserves LCR buffer and stabilizes core deposits
Higher term funding cost, lower liquidity volatility	Greater rate sensitivity, but flexible repricing

Decision: a **hybrid approach** combining partial covered bond issuance and selective retail repricing achieves both stability and profitability objectives.

4.10 Governance, Policies, and Regulatory Expectations

Effective liquidity risk management is underpinned by a strong **governance framework** integrating Board oversight, ALCO decision-making, and risk reporting.

According to **EBA Guidelines on Internal Liquidity Adequacy Assessment Process (ILAAP, 2019)**, key elements include:

- Defined **Liquidity Risk Appetite (LRA)** and tolerance limits;
- Periodic **ILAAP review** integrated with ICAAP;
- **Independent validation** of models and assumptions;
- Regular **Board reporting** and supervisory dialogue.

Supervisors emphasize consistency between liquidity, funding, and interest rate risk governance, particularly under **SREP (Supervisory Review and Evaluation Process)** assessments.

4.11 Emerging Themes: Liquidity in a Digital and ESG Context

Two structural trends reshape the liquidity landscape:

1. Digital Transformation and Deposit Volatility

Mobile banking accelerates deposit mobility and amplifies “digital bank run” dynamics. ALM models must adapt behavioral parameters to shorter decision lags and social contagion effects.

2. ESG and Green Funding Instruments

The rise of **Green Bonds** and **Sustainability-Linked Loans** introduces a new funding dimension — aligning liquidity profiles with sustainability targets. Investors increasingly reward institutions with transparent ESG-linked funding strategies, creating an intersection between *liquidity optimization* and *sustainable finance*.

4.12 References

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Chapter 5 – Balance Sheet Optimization and Strategic ALM

5.1 Introduction

In modern banking, the **balance sheet is the core reflection of a bank's risk–return profile**. Each asset, liability, and off-balance-sheet item contributes to the institution's structural positioning toward interest rate, liquidity, and credit risks.

The **Asset and Liability Management (ALM)** function thus serves not only as a risk control mechanism but as a *strategic engine* that shapes the bank's profitability, capital efficiency, and resilience.

Balance sheet optimization seeks to identify the configuration of assets and liabilities that maximizes shareholder value—typically measured in terms of *risk-adjusted return on capital (RAROC)* or *economic value added (EVA)*—while remaining compliant with internal and regulatory constraints such as capital, liquidity, and interest rate limits.

This chapter explores the analytical framework, quantitative tools, and governance processes that underpin balance sheet optimization in the ALM context, culminating in an integrated model of *Strategic ALM*.

5.2 Objectives of Strategic ALM

Strategic ALM integrates financial, regulatory, and economic dimensions into a coherent decision-making framework.

Its objectives can be summarized as follows:

1. **Optimize the structural balance sheet mix** to achieve a target risk–return trade-off;
2. **Ensure consistency with the bank's Risk Appetite Framework (RAF)** and regulatory requirements (ICAAP, ILAAP, IRRBB);
3. **Stabilize earnings and capital** under plausible macroeconomic and rate scenarios;
4. **Support strategic planning** by aligning funding, investment, and hedging strategies with medium-term business objectives;
5. **Facilitate ALCO decision-making** through quantitative dashboards and scenario simulations.

Ultimately, Strategic ALM transforms the balance sheet from a static construct into a dynamic portfolio, managed with the same sophistication as a financial investment fund but constrained by regulatory and fiduciary principles.

5.3 The Optimization Problem in ALM

At its core, balance sheet optimization can be represented as a **multi-objective constrained optimization problem**.

A simplified formalization is:

$$\begin{aligned} & \max_{\mathbf{x}} \quad R(\mathbf{x}) - \lambda \cdot \sigma(\mathbf{x}) \\ \text{subject to: } & \begin{cases} C(\mathbf{x}) \leq C_{min} & (\text{Capital constraints}) \\ LCR(\mathbf{x}) \geq 100\% & (\text{Liquidity coverage}) \\ NSFR(\mathbf{x}) \geq 100\% & (\text{Stable funding}) \\ IRRBB(\mathbf{x}) \leq \text{limit} & (\text{Interest rate risk limits}) \\ \sum_i x_i = 1 & (\text{Balance sheet identity}) \end{cases} \end{aligned}$$

Where:

- $R(\mathbf{x})$ = expected return of balance sheet configuration $\mathbf{x}$;
- $\sigma(\mathbf{x})$ = volatility (risk measure, e.g. NII or EVE volatility);
- λ = risk aversion coefficient;
- $C(\mathbf{x})$, $LCR(\mathbf{x})$, etc. are nonlinear constraint functions.

In practice, optimization models combine *stochastic simulations* (for interest rate, spread, and liquidity shocks) with *linear programming* or *Monte Carlo optimization techniques* to find efficient portfolios of assets and liabilities.

5.4 The Efficient Frontier in ALM

The concept of the **Efficient Frontier**, derived from Modern Portfolio Theory (Markowitz, 1952), can be adapted to the banking context.

Here, the objective is not to maximize returns for a given asset portfolio, but to **maximize Net Interest Income (NII) or Return on Assets (ROA) for a given level of Interest Rate Risk (IRR)**.

Example representation:

Bank	NII / Total Assets	Δ NII / Tier 1	Positioning
Bank A	2.0%	4.5%	Efficient
Bank B	1.6%	6.0%	Over-risked
Bank C	1.9%	3.8%	Balanced
Bank D	1.3%	2.5%	Under-risked

Plotting these data points in a two-dimensional space (Return vs. Risk) yields the **ALM Efficient Frontier**, identifying which configurations deliver the best trade-off between profitability and sensitivity.

Banks positioned below the frontier are *inefficient*, either taking too much risk for a given return or earning too little for their risk exposure.

This framework is instrumental for **ALCO decisions**, allowing management to benchmark its structural performance against peers or internal targets.

5.5 Risk–Return Trade-offs: NII, EVE, and Capital Sensitivity

ALM optimization requires the simultaneous management of three key sensitivities:

Dimension	Measure	Strategic Objective
Short-term performance	Δ NII	Stabilize earnings under rate shocks
Long-term value	Δ EVE / Tier 1	Protect capital from rate shifts
Capital efficiency	RWA / Tier 1	Maximize risk-adjusted return

The interplay between these factors defines the **risk–return map** of the balance sheet.

For instance, extending asset duration (longer fixed-rate loans or sovereign bonds) typically increases NII under rising rates but reduces EVE sensitivity and consumes more capital under IRRBB rules.

Thus, ALM optimization involves navigating *regulatory capital*, *liquidity buffers*, and *rate exposure* simultaneously.

5.6 Integration with Capital and Liquidity Planning

Effective balance sheet optimization requires consistent integration across:

1. **Capital Planning (ICAAP)** – aligning IRRBB and market risk exposures with available capital buffers;
2. **Liquidity Planning (ILAAP)** – ensuring funding stability and LCR/NSFR compliance under optimized scenarios;
3. **Strategic Planning** – ensuring the ALM structure supports long-term profitability and growth.

This integration is operationalized through a **Strategic ALM Model (SALM)**—a multi-year projection tool that simulates NII, EVE, capital, and liquidity under macro scenarios, providing quantitative input to the bank’s *Strategic Plan* and *Recovery Plan*.

5.7 Quantitative Tools and Scenario Analysis

Modern ALM optimization leverages advanced analytical tools, including:

- **Dynamic Balance Sheet Models (DBSM):** simulate future balance sheet evolution under macroeconomic scenarios (GDP, inflation, rates).
- **Interest Rate Scenario Engines:** generate stochastic rate paths consistent with yield curve dynamics and volatility structures (e.g., Nelson–Siegel models).
- **ALM Optimizers:** employ linear or stochastic programming to maximize NII subject to regulatory constraints.
- **Sensitivity Dashboards:** visualize Δ NII, Δ EVE, and risk-weighted asset changes under base and stress conditions.

These tools form the quantitative backbone of **data-driven ALCO decisions**, linking modeling outputs directly to strategic steering.

5.8 Governance and Decision Framework: The ALCO Role

The **Asset & Liability Committee (ALCO)** is the decision-making body responsible for steering the balance sheet within the Risk Appetite and regulatory limits.

ALCO responsibilities include:

- Reviewing NII/EVE sensitivity results and approving hedging actions;
- Balancing short-term profitability against long-term stability;
- Approving funding and investment strategies;
- Coordinating with Treasury, Risk Management, and Finance functions;
- Monitoring regulatory metrics (LCR, NSFR, IRRBB).

ALCO effectiveness depends on **timely, integrated information** from ALM dashboards, ensuring strategic alignment between *risk measurement* and *business objectives*.

5.9 Practical Case: Strategic Repositioning under Rising Rates

Consider a commercial bank with:

- NII/Tot. Assets = 1.7%;
- $\Delta\text{NII}/\text{Tier 1} = 7.0\%$;
- Duration gap = +0.25 years;
- SOT (Supervisory Outlier Test) NII = -7.5%.

Facing a rising rate environment, the ALCO explores two optimization strategies:

Strategy	Description	Expected Effect
1. Asset Reallocation	Buy €2bn of 2-year sovereign bonds (FVOCI)	+0.15% NII, improves $\Delta\text{NII}/\text{Tier 1}$ to 6.3%
2. Derivative Strategy	Enter receiver IRS (€2bn, 2Y)	Lower ΔNII volatility, reduce EVE sensitivity
3. Funding Adjustment	Increase term funding share from 15% → 25%	Improves NSFR to 115%, slight NIM compression

Result: The combined strategy enhances stability (lower ΔNII volatility) while preserving profitability and improving liquidity resilience.

This demonstrates how *ALM optimization* translates into tactical adjustments aligned with strategic goals.

5.10 Challenges and Future Directions

The increasing complexity of financial systems introduces new challenges for ALM optimization:

1. **Regulatory Interaction:** overlapping constraints (IRRBB, LCR, NSFR, TLAC, MREL) complicate optimization objectives.
2. **Behavioral Uncertainty:** deposit and loan behavior under stress remains difficult to model accurately.
3. **Model Risk:** excessive reliance on deterministic assumptions can lead to misaligned hedging decisions.

4. **ESG and Green Transition:** balancing sustainability targets with risk–return optimization introduces new decision layers.
5. **AI and Machine Learning Integration:** emerging techniques can enhance scenario calibration and behavioral modelling but raise transparency concerns.

The next frontier in ALM optimization lies in the development of **Integrated Digital ALM Platforms**, capable of real-time scenario simulation and decision support across Treasury, Risk, and Finance.

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Chapter 6 – Interest Rate Risk in the Banking Book (IRRBB)

6.1 Introduction

Interest Rate Risk in the Banking Book (IRRBB) represents one of the most critical components of a bank's overall risk profile.

Unlike trading book risk—typically marked to market and actively hedged—IRRBB originates from the *structural mismatch* between interest-sensitive assets and liabilities in the *banking book*, which are generally held at amortized cost.

Fluctuations in interest rates affect both:

1. **Earnings** — through changes in *Net Interest Income (NII)*, and
2. **Economic value** — through the impact on the *Economic Value of Equity (EVE)*.

Managing IRRBB is therefore essential to preserving both short-term profitability and long-term capital stability.

Under **Basel standards** and the **EBA Guidelines on IRRBB (2021)**, banks must identify, measure, monitor, and control their interest rate exposures, ensuring that they remain consistent with their **Risk Appetite Framework (RAF)** and capital adequacy requirements (*ICAAP*).

6.2 Components of IRRBB

According to the **Basel Committee on Banking Supervision (BCBS, 2016)**, IRRBB can be decomposed into several sub-components, each reflecting a different source of rate sensitivity:

Type of IRRBB	Definition	Typical Drivers
Gap risk	Arises from mismatched repricing dates between assets and liabilities	Fixed vs. variable rate mix, maturity transformation
Basis risk	Arises from imperfect correlation between different rate indices	Euribor vs. EONIA/SOFR, different repricing bases
Yield curve risk	Arises from changes in the slope or shape of the yield curve	Steepening/flattening scenarios
Option risk	Arises from embedded options in products	Loan prepayments, deposit early withdrawals
Convexity risk	Arises from non-linear sensitivity to large rate movements	Mortgage portfolios, callable bonds

Banks must quantify and manage all these components using both **economic value** and **earnings-based** approaches.

6.3 Measurement Approaches: EVE and NII

(a) Economic Value of Equity (EVE)

The **EVE** represents the present value of expected future cash flows from assets, liabilities, and off-balance-sheet items.

Changes in EVE under parallel or non-parallel rate shocks indicate the *structural sensitivity* of the bank's capital to interest rate movements.

$$\Delta EVE = EVE_{shock} - EVE_{base}$$

EVE is particularly relevant for assessing **long-term interest rate exposure**. A large negative ΔEVE implies that the bank's economic value—and therefore its capital—is eroded by rate movements.

(b) Net Interest Income (NII)

The ΔNII metric measures the variation in projected NII over a one-year horizon under alternative rate scenarios (e.g., ± 200 basis points).

$$\Delta NII = NII_{shock} - NII_{base}$$

Unlike EVE, NII focuses on the **short-term earnings impact** of rate shocks. A negative ΔNII under a rate increase typically signals that the bank has more rate-sensitive liabilities than assets—i.e., a *liability-sensitive balance sheet*.

6.4 Regulatory Framework

Under **Basel III** and **EBA Guidelines (EBA/GL/2021/10)**, banks are required to:

1. Measure ΔEVE and ΔNII under six standard rate shock scenarios (parallel up/down, steepener, flattener, short up, long up).
2. Report results both in absolute (€) and relative (% of Tier 1 capital) terms.
3. Identify “outlier banks” under the **Standardized Outlier Test (SOT)**:
 - **EVE SOT threshold**: $\Delta EVE / \text{Tier 1} \leq -15\%$
 - **NII SOT threshold**: $\Delta NII / \text{Tier 1} \leq -5\%$ (non-binding but monitored)
4. Integrate IRRBB into ICAAP and stress testing frameworks.

These requirements aim to ensure **comparability, transparency, and supervisory consistency** across European institutions.

6.5 Duration Gap and Sensitivity Analysis

The **duration gap (DGAP)** is a fundamental measure of a bank’s exposure to interest rate risk, reflecting the weighted average repricing mismatch between assets and liabilities:

$$DGAP = D_A - \frac{L}{A} D_L$$

where:

- D_A = duration of assets,
- D_L = duration of liabilities,
- L/A = leverage ratio of liabilities to assets.

The change in EVE can be approximated using the duration gap:

$$\Delta EVE \approx -DGAP \cdot A \cdot \Delta y$$

where Δy is the interest rate shock.

Interpretation:

- A *positive* DGAP → asset-sensitive → benefits from rising rates (EVE down, NII up).
- A *negative* DGAP → liability-sensitive → benefits from falling rates (EVE up, NII down).

6.6 Advanced Modelling: Optionality and Behavioral Adjustments

In practice, IRRBB measurement must incorporate **behavioural assumptions** about:

- **Prepayment rates** on fixed-rate loans;
- **Early redemption** of term deposits;
- **Non-maturing deposit (NMD)** stability;
- **Loan repricing elasticity** to policy rates.

Banks use models such as:

- **Option-adjusted duration (OAD)** for portfolios with embedded options;
- **Replication portfolios** to model NMDs;
- **Monte Carlo simulations** for stochastic rate paths and optionality.

The inclusion of **convexity** and **Vega effects**—sensitivity to rate volatility—is increasingly important in stress scenarios, as shown during the 2023 U.S. regional bank crisis (Jiang et al., 2023).

6.7 Strategic Implications and ALM Actions

The ALCO uses IRRBB metrics to make **strategic decisions** in areas such as:

Action Area	Objective	ALM Levers
Repricing profile	Adjust duration and sensitivity	Asset mix, term funding, swaps
Hedging	Mitigate EVE/NII volatility	IRS, swaptions, macro-FVH
Investment portfolio	Optimize yield vs. sensitivity	HTC vs. FVOCI allocation
Funding	Stabilize costs, manage optionality	Retail term deposits, covered bonds
Capital management	Control IRRBB capital absorption	ICAAP integration, risk transfer

These actions must be **quantitatively supported** through scenario simulations and consistently documented for accounting and regulatory purposes (IFRS 9 + EBA GL/2021/10).

6.8 Example: Parallel Rate Shock Scenario

Scenario	Rate Shock (bps)	Δ EVE (€m)	Δ EVE / Tier 1	Δ NII (€m)	Δ NII / Tier 1	Interpretation
Base	0	0	0%	0	0%	–
Parallel Up	+200	–45	–12%	+22	+5.8%	Asset-sensitive
Parallel Down	–200	+40	+10%	–24	–6.3%	Earnings exposed
Steepener	Short +100 / Long –100	–20	–5%	+10	+2.5%	Front-end exposure
Flattener	Short –100 / Long +100	–30	–8%	–8	–2.1%	Long-end risk

Result:

The bank exhibits moderate *EVE sensitivity* and positive *earnings sensitivity*. ALCO may consider partial **duration reduction** or **macro-hedging** to reduce capital volatility while preserving NII upside.

6.9 Governance, Reporting, and Integration

Effective IRRBB governance relies on:

- **ALM policy and limits framework** (Δ EVE, Δ NII, DGAP thresholds);
- **Independent validation** of IRRBB models;
- **Integration with ICAAP/ILAAP and stress testing**;
- **Comprehensive reporting** to ALCO, Board, and Supervisors;
- **Documentation for Hedge Accounting** under IFRS 9 (especially for macro-FVH).

Transparency and traceability are critical for supervisory reviews (SREP), where qualitative governance assessments weigh as much as quantitative outcomes.

6.10 Future Trends in IRRBB Management

Emerging challenges and trends include:

1. **Rate Volatility and Convexity Risk:** growing need to assess Vega and gamma exposure.
2. **Behavioral Modelling Enhancements:** use of AI/ML to model NMDs and prepayments dynamically.
3. **Integration with ESG Risk:** aligning IRRBB with sustainability-linked funding and green lending.
4. **Macro-FVH Expansion:** adoption of hedge accounting for structural portfolios.

5. **Digital ALM Platforms:** integration of real-time analytics and scenario simulation capabilities.

The future of IRRBB management lies in combining **quantitative sophistication** with **strategic adaptability**—ensuring that banks can preserve earnings stability while navigating complex, volatile rate environments.

6.11 References

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Chapter 7 – Behavioral Modeling and Non-Maturing Deposits (NMDs)

7.1 Introduction

Among all the elements of interest rate risk management, the treatment of **non-maturing deposits (NMDs)** stands out as one of the most challenging.

NMDs—such as demand deposits, current accounts, and savings accounts—lack a contractual maturity. They can be withdrawn at any time, yet historically display high stability and predictable behavioural patterns.

This dual nature—*legally short-term but behaviourally long-term*—creates a modeling dilemma: how should banks assign a duration to an instrument that, in theory, can reprice daily, but in practice behaves like a multi-year liability?

The answer lies in **behavioural modeling**: a set of quantitative and econometric techniques that translate observed depositor behaviour into stable, risk-sensitive parameters for **interest rate risk, liquidity management, and fund transfer pricing (FTP)**.

Understanding and modeling NMDs accurately is essential for:

- **Interest Rate Risk in the Banking Book (IRRBB)** quantification (EVE and NII sensitivity);
- **Liquidity Coverage Ratio (LCR)** and **Net Stable Funding Ratio (NSFR)** projections;
- **Hedge Accounting** effectiveness (especially macro-fair value hedges);
- **Strategic ALM positioning** and funding mix optimization.

7.2 Nature and Classification of Non-Maturing Deposits

The European Banking Authority (EBA, *Guidelines on IRRBB and CSRBB*, 2021) distinguishes between two key categories of NMDs:

Category	Examples	Typical Behavioural Features
Retail NMDs	Current accounts, savings accounts held by households	Stable, low rate sensitivity, slow reaction to market rate changes
Corporate NMDs	Operational deposits of businesses, public entities	More volatile, faster repricing, sensitive to opportunity costs

Retail deposits typically exhibit **longer behavioural durations** and **lower elasticity** to policy rates than corporate deposits.

Conversely, institutional and wholesale deposits are more rate-sensitive and require cautious modeling, especially under stress conditions.

7.3 Behavioral Assumptions and Stability Analysis

To model NMDs, banks must estimate three key behavioural parameters:

1. **Core vs. Non-Core Portion:**

The *core* portion represents stable deposits that are expected to remain even under adverse conditions.

The *non-core* portion is volatile and rate-sensitive, often withdrawn when market yields rise.

$$\text{Core Ratio} = \frac{\text{Stable Balances}}{\text{Total NMDs}}$$

2. **Average Life (Behavioural Duration):**

This is the weighted average time until repricing, reflecting customer stickiness. Empirical studies show retail NMDs often have behavioural durations between **2 and 5 years**.

3. **Rate Sensitivity (Elasticity):**

Measured as the deposit rate's responsiveness to changes in market rates:

$$\text{Elasticity} = \frac{\Delta r_{\text{deposit}}}{\Delta r_{\text{market}}}$$

Elasticities typically range from **0.1–0.3 for retail** and **0.5–0.8 for corporate** accounts.

7.4 Modeling Techniques

Modern ALM functions use a mix of **statistical**, **econometric**, and **machine learning** techniques to estimate behavioural parameters.

The most widely adopted methodologies include:

(a) Time Series and Regression Models

Linear regressions or ARIMA models capture the relationship between changes in deposit balances/rates and market variables (e.g. Euribor, GDP, unemployment).

(b) Survival and Attrition Models

Used to estimate expected life and withdrawal probabilities of deposits. They model customer behaviour as a time-to-event process (e.g., Cox proportional hazards models).

(c) Replication Portfolio Approach

Perhaps the most operationally relevant for ALM, this method models the NMD portfolio as a synthetic combination of fixed-rate instruments that replicate its duration and repricing characteristics.

$$\text{Replicating Portfolio Value} = \sum_i w_i \times PV_i$$

where w_i are portfolio weights assigned to benchmark bonds or swaps of different maturities.

(d) Machine Learning Extensions

Neural networks and random forests can capture non-linear relationships between deposit behaviour and macroeconomic drivers (inflation, unemployment, consumer confidence). However, regulatory acceptance remains cautious, emphasizing **interpretability and backtesting**.

7.5 The Replication Portfolio Framework

The **replication portfolio** remains the gold standard under EBA and Basel IRRBB guidance. It allows ALM desks to translate uncertain behavioural liabilities into *market-based instruments* that can be risk-managed and hedged.

Steps in Building a Replication Portfolio:

1. **Segment deposits** (e.g., retail current, retail savings, corporate).
2. **Estimate behavioural duration and elasticity** for each segment.
3. **Construct a synthetic portfolio** of fixed-rate assets or swaps that replicate the expected cash flow pattern.
4. **Test performance** under multiple rate and liquidity scenarios.
5. **Update periodically** based on observed depositor behaviour.

This approach also enables accounting alignment under **macro fair value hedge accounting (FVH)**—since the replicated exposures can be designated as hedged items against interest rate swaps.

7.6 Stress Testing and Scenario Analysis

Behavioural models must be complemented by **stress scenarios** to evaluate deposit stability under extreme conditions.

Stress Driver	Description	Impact on NMDs
Rate Shocks	Sudden rate hikes or cuts	Core ratio declines as customers reallocate to term deposits
Liquidity Stress	Market disruption or loss of confidence	Spike in withdrawals, especially from corporate deposits
Reputational Event	Negative news or credit downgrade	Accelerated outflows
Digital Channel Dynamics	Faster transfer mechanisms increase volatility	Shorter effective duration

For regulatory purposes, the **Basel III liquidity framework** (LCR, NSFR) provides conservative stability assumptions that serve as lower bounds for model validation.

7.7 Integration with IRRBB and Liquidity Metrics

Behavioural NMD modeling influences both **interest rate** and **liquidity risk** measures:

Metric	Dependence on NMD Modeling	Effect of Higher Duration/Core Stability
ΔEVE	Longer behavioural duration increases EVE sensitivity	Higher capital volatility
ΔNII	Greater stickiness stabilizes NII under rate shifts	Lower earnings volatility
LCR	Core deposits treated as stable funding	Improves LCR denominator
NSFR	Long-term stable deposits improve stable funding ratio	Enhances structural liquidity

Hence, NMD assumptions directly shape the bank's **ALCO decisions** and **ICAAP-ILAAP coherence**.

7.8 Example: Behavioral Replication and Sensitivity

Below is an illustrative example of how behavioural modeling alters balance sheet sensitivity:

Portfolio Segment	Nominal (€bn)	Core Ratio	Behavioral Duration (yrs)	Elasticity	Modeled IRRBB Impact (ΔEVE, €m)
Retail Current Accounts	5.0	85%	3.5	0.2	–40
Retail Savings	2.0	70%	2.8	0.3	–25
Corporate Operational	3.0	55%	1.0	0.6	–15
Total NMDs	10.0	74% avg.	2.6 yrs avg.	–	–80

When behavioural stability is re-estimated (e.g., post-COVID), duration may drop by 30–40%, leading to an improvement in ΔEVE but an increase in ΔNII volatility—showing the **trade-off between value and earnings stability**.

7.9 Governance and Model Validation

Supervisors expect banks to maintain robust model governance, including:

- **Independent validation** of behavioural assumptions;
- **Regular backtesting** vs. actual deposit retention rates;
- **Change management and recalibration procedures**;
- **Documentation and transparency** of modeling techniques;
- **Integration into ICAAP and SREP reporting**.

EBA explicitly requires that behavioural parameters “reflect observed, stable customer behaviour and be consistent with macroeconomic conditions” (*EBA/GL/2021/10, para. 142–150*).

7.10 Strategic Implications for ALM

The strategic relevance of behavioural modeling extends beyond compliance. It shapes:

1. **FTP policies** (internal cost of funds allocation);
2. **Pricing of retail products** (through elasticity and duration metrics);
3. **Hedging strategy calibration** (e.g., IRS tenor selection);
4. **Liquidity buffer optimization**;
5. **Capital planning and stress scenario design**.

Banks that master behavioural modeling can achieve a competitive advantage through:

- More accurate margin forecasting;

- Lower funding cost volatility;
- Optimized hedge efficiency;
- Improved strategic decision-making in ALCO forums.

7.11 Conclusion

Behavioral modeling of non-maturing deposits sits at the intersection of *quantitative finance*, *behavioural economics*, and *strategic ALM management*.

An accurate representation of deposit behaviour allows banks to:

- Align interest rate and liquidity risk measurement;
- Support hedge accounting designations under IFRS 9;
- Anticipate customer reactions to monetary policy changes;
- Strengthen resilience under stress conditions.

In an era of increasing digitalization and rate volatility, the challenge is to build behavioural models that are both **data-rich and adaptive**, integrating real-time behavioural signals with robust econometric foundations.

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Chapter 8 – Liquidity Risk Management and Funding Strategy

8.1 Introduction

Liquidity risk represents the potential inability of a financial institution to meet its obligations as they come due, without incurring unacceptable losses.

It is fundamentally distinct from solvency risk: while capital protects against *losses*, liquidity protects against *timing mismatches*.

The 2008 Global Financial Crisis revealed that **illiquidity can destroy otherwise solvent institutions** within days. Consequently, international regulation—particularly the **Basel III framework**—introduced quantitative liquidity standards (LCR, NSFR) and strengthened qualitative expectations for **Liquidity Risk Management Frameworks (LRMFs)**.

From an **Asset and Liability Management (ALM)** perspective, liquidity management complements interest rate and funding strategies, ensuring that the bank maintains a balanced, stable, and cost-efficient funding structure under both normal and stressed conditions.

8.2 Types of Liquidity Risk

Liquidity risk manifests through multiple channels. The main categories are:

Type	Definition	Examples / Indicators
Funding Liquidity Risk	Inability to obtain funding to meet obligations	Deposit outflows, loss of market access
Market Liquidity Risk	Inability to sell assets without significant price concessions	Bond market illiquidity, repo haircuts
Intraday Liquidity Risk	Short-term liquidity gaps within a single day	Payment system delays, collateral transfers
Contingent Liquidity Risk	Potential future liquidity needs from off-balance-sheet exposures	Loan commitments, derivatives margin calls

These risks are closely interconnected and require an integrated **Liquidity and Funding Risk Management Framework (LFRMF)**.

8.3 The Liquidity Coverage Ratio (LCR)

The **Liquidity Coverage Ratio (LCR)** ensures that banks hold an adequate buffer of high-quality liquid assets (HQLA) to survive a severe 30-day stress scenario.

$$\text{LCR} = \frac{\text{Stock of HQLA}}{\text{Total Net Cash Outflows over 30 days}} \geq 100\%$$

Key components:

- **HQLA:** Cash, central bank reserves, government bonds, and other high-quality securities subject to haircuts.
- **Outflows:** Expected withdrawals from retail, corporate, and wholesale deposits.
- **Inflows:** Contractual cash inflows capped at 75% of outflows.

The LCR is calibrated to ensure that banks can withstand *idiosyncratic* and *market-wide* stress scenarios without external support.

8.4 The Net Stable Funding Ratio (NSFR)

The **NSFR** complements the LCR by promoting stable funding over a one-year horizon.

$$\text{NSFR} = \frac{\text{Available Stable Funding (ASF)}}{\text{Required Stable Funding (RSF)}} \geq 100\%$$

- **ASF:** Equity, long-term liabilities, and stable deposits.
- **RSF:** Assets weighted by liquidity and maturity characteristics.

NSFR ensures that longer-term assets are funded by sufficiently stable liabilities, thus limiting excessive maturity transformation.

8.5 Liquidity Stress Testing

Stress testing forms the **core of the internal liquidity risk framework**.

Banks must design stress scenarios capturing institution-specific vulnerabilities and systemic crises.

Stress Type	Drivers	Expected Effects
Idiosyncratic	Credit downgrade, reputational loss	Sudden retail/corporate outflows
Market-wide	Financial crisis, policy tightening	Funding market closure, collateral devaluation
Combined	Interaction of both	Liquidity spiral, rising haircuts, asset fire sales

Stress results feed into **contingency funding plans (CFPs)**, which outline mitigation measures and emergency funding sources.

8.6 The Role of ALM in Liquidity Management

ALM acts as the **bridge between liquidity and interest rate management**. Its strategic objectives include:

1. **Maintaining structural liquidity** through an optimal asset-liability maturity profile.
2. **Ensuring funding diversification** across instruments, tenors, and counterparties.
3. **Optimizing the liquidity buffer** composition to balance cost and usability.
4. **Integrating IRRBB and liquidity analytics** in unified scenario frameworks.
5. **Supporting regulatory ratios** (LCR, NSFR) while sustaining profitability.

The **Asset and Liability Committee (ALCO)** defines limits, monitors gaps, and approves funding and investment actions consistent with the bank's risk appetite.

8.7 Funding Strategy and Composition

An effective **funding strategy** relies on diversification and stability.

The mix between retail deposits, wholesale funding, and capital market instruments defines both **liquidity resilience** and **interest rate sensitivity**.

Funding Source	Characteristics	Advantages	Challenges
Retail Deposits	Stable, low-cost, behaviourally sticky	Improves LCR/NSFR	Limited scalability
Wholesale Funding	Interbank, repos, senior unsecured	Large volumes, flexibility	Sensitive to market confidence
Covered Bonds / Securitizations	Asset-backed	Collateralized, long-term	Complexity, regulatory caps
Central Bank Funding	TLTROs, standing facilities	Reliable under stress	Temporary, policy-dependent
Capital Markets	Senior/non-preferred debt, Tier 2	Improves term structure	Market access volatility

8.8 Liquidity Gap and Maturity Mismatch Analysis

Liquidity risk is monitored through **maturity gap profiles**, displaying net cumulative cash flows by time bucket.

Time Bucket	Cumulative Gap (€bn)	Interpretation
Overnight	−2.0	Short-term dependency on interbank lines
1–7 days	−1.5	LCR critical zone
1–3 months	−0.5	Neutral funding
3–12 months	+0.8	Medium-term surplus
>1 year	+3.0	Structural funding buffer

ALM aims to maintain a **negative gap only in short horizons**, offset by the HQLA buffer, while preserving a **positive long-term structural position**.

8.9 Integration of Liquidity and Interest Rate Risk

Liquidity and IRRBB are inherently linked:

- **Short-term funding increases rate sensitivity**, as repricing occurs faster.
- **Long-term fixed funding stabilizes earnings**, but raises liquidity costs.
- **Optionality in deposits** creates both liquidity and rate uncertainty.

ALM must therefore evaluate **combined shocks**—e.g., rate hikes coinciding with funding stress—through integrated *EVE-NII-LCR* simulations.

$$EVE_{adj} = f(\Delta y, \text{Liquidity Gap}, \text{Funding Cost Elasticity})$$

8.10 Example: Funding Optimization under Stress

Scenario	Funding Action	LCR Impact	NII Impact	Strategic Consideration
Baseline	Balanced retail & wholesale mix	120%	+0	Stable funding structure
Stress 1: Rate hike +100 bps	Shift to longer-term retail funding	+10%	−15m	Increases LCR, reduces margin
Stress 2: Market freeze	Use ECB facilities and repo HQLA	+20%	−10m	Preserves liquidity, increases collateral usage
Stress 3: Deposit outflow 5%	Issue covered bonds	+15%	−8m	Strengthens NSFR, increases duration

This demonstrates the **trade-off between liquidity protection and earnings stability**, a central theme in modern ALM.

8.11 Contingency Funding Plan (CFP)

Every institution must maintain a **CFP** detailing:

- Early warning indicators (market spreads, liquidity ratios);
- Escalation procedures and communication lines;
- Pre-arranged collateral and repo lines;
- Central bank facility access;
- Recovery options and prioritization.

The CFP is the operational backbone of **liquidity crisis management**, linking strategic planning with tactical response.

8.12 Governance and Supervision

Supervisory expectations (EBA/GL/2018/02) require:

- **Board-level oversight** of liquidity risk appetite;
- **Integrated ICAAP/ILAAP documentation;**

- **Independent liquidity risk monitoring;**
- **Internal pricing through FTP frameworks;**
- **Regular validation and backtesting** of liquidity assumptions.

The alignment between **liquidity strategy** and **capital planning** ensures consistency within the bank's *overall risk governance architecture*.

8.13 Emerging Trends in Liquidity Management

Future liquidity management will be shaped by:

1. **Digital transformation** — real-time liquidity dashboards, intraday analytics.
2. **ESG-linked funding** — green bonds and sustainable deposits influencing funding mix.
3. **Central bank digital currencies (CBDCs)** — potential new liquidity channels.
4. **Integrated stress testing** — joint simulation of market, liquidity, and interest rate risk.
5. **Behavioural analytics** — AI-based monitoring of depositor behaviour and funding flows.

These evolutions redefine the boundaries of ALM, positioning liquidity as both a risk and a strategic resource.

8.14 Conclusion

Liquidity management is no longer a defensive discipline; it is a **strategic determinant of resilience and profitability**.

An optimized funding strategy allows banks to:

- Safeguard stability under stress,
- Support business growth,
- Manage interest rate sensitivity proactively, and
- Comply efficiently with regulatory standards.

In modern banking, **liquidity, funding, and interest rate risk** form an integrated triad: mastering their interaction defines the competitive edge of sophisticated ALM functions.

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Chapter 9 – Funds Transfer Pricing (FTP) and Internal Capital Allocation

9.1 Introduction

Funds Transfer Pricing (FTP) lies at the heart of the **Asset and Liability Management (ALM)** framework.

It is the mechanism through which the **costs, benefits, and risks of funding and liquidity** are internally priced and transferred across a bank's business units.

While regulatory frameworks (Basel III, EBA SREP) require that risks be properly identified, measured, and capitalized, FTP ensures that these risks are **economically internalized**—that is, correctly reflected in internal performance metrics and product pricing.

An effective FTP system serves three core purposes:

1. **Pricing:** ensuring that internal product rates reflect funding, liquidity, and interest rate risk costs;
2. **Performance Measurement:** aligning profitability indicators with risk-adjusted returns (RAROC, RORWA);
3. **Strategic Steering:** supporting ALCO in resource allocation and balance sheet optimization.

9.2 The Rationale for FTP

In a modern, diversified bank, different divisions—Retail, Corporate, Treasury—generate assets and liabilities with varying maturities, optionalities, and behavioural characteristics. Without FTP, product-level profitability would be distorted by central funding and interest rate movements.

FTP acts as an *internal market mechanism*, ensuring:

- **Neutrality to funding structure:** business lines are evaluated independently of the bank's external funding mix;
- **Alignment with ALM strategy:** promoting behaviours consistent with the bank's risk appetite (e.g., stable deposits, long-term funding);
- **Transparency and accountability:** business units understand the cost of liquidity and rate risk embedded in their portfolios.

9.3 FTP Framework Structure

A comprehensive FTP framework typically includes three interacting components:

Component	Purpose	Managed by
Base Transfer Rate (BTR)	Reflects the cost of term funding or investment for given maturities	ALM / Treasury
Liquidity Premium (LP)	Captures cost of illiquidity or funding asymmetry	ALM / Risk Management
Risk Adjustment (RA)	Reflects optionality, behavioural, or embedded risk	ALM / Product Units

The sum of these components determines the **Internal Transfer Rate (ITR)**:

$$ITR = BTR + LP + RA$$

Each product's internal margin is then calculated as:

$$\text{Internal Margin} = \text{Client Rate} - \text{ITR}$$

9.4 Base Transfer Rate (BTR)

The BTR represents the **risk-free term structure** of interest rates at which the bank can fund itself or invest.

It is generally derived from the **swap curve** or **bank-specific funding curve**, and should reflect:

- The *currency* and *tenor* of the transaction;
- The *repricing profile* of the product (fixed or floating);
- The *hedging horizon* defined by ALM.

Example:

Tenor	Reference Rate (EUR Swap)	Base Transfer Rate (bps)
3 months	3M Euribor	0
1 year	3M Euribor + 20	+20
3 years	3M Euribor + 40	+40
5 years	3M Euribor + 60	+60

ALM updates the BTR curve periodically, ensuring internal consistency between product pricing and external market rates.

9.5 Liquidity Premium (LP)

The liquidity premium accounts for **structural funding imbalances** between assets and liabilities.

It reflects the cost (or benefit) of obtaining stable funding beyond the horizon of liquid markets.

$$LP = f(\text{Funding Gap, Market Liquidity, LCR/NSFR Buffer})$$

Typical considerations include:

- Market liquidity conditions (repo spreads, bond bid-ask);
- The bank's funding plan and NSFR position;
- Behavioural stability of deposits;
- Regulatory liquidity buffers.

Product	Liquidity Premium (bps)
Retail mortgage (10Y)	+30
SME loan (5Y)	+20
Corporate credit line (undrawn)	+40
Retail deposit (core, stable)	-10

Liquidity premiums are dynamic and may vary according to **ALCO decisions** and **stress testing outcomes**.

9.6 Risk Adjustment (RA)

Risk adjustments capture **embedded optionalities and behavioural uncertainties**, such as:

- Prepayment risk on loans and mortgages;
- Early withdrawal options on deposits;
- Rate floors and caps on floating instruments.

These adjustments are typically estimated using **option-adjusted spread (OAS)** models or behavioural simulations, and ensure that business units bear the economic cost of embedded risks they generate.

9.7 FTP for Product Segments

FTP application varies by product family:

Product Segment	FTP Design	Key Considerations
Fixed-rate loans	Transfer fixed BTR for loan maturity	Prepayment optionality
Floating-rate loans	Transfer rolling short-term BTR	Basis risk
Term deposits	Transfer matching BTR	Early redemption
Current accounts (NMDs)	Apply behavioural FTP rate based on replication duration	Stability, core portion
Off-balance-sheet lines	Transfer liquidity cost only	Drawdown probability

FTP thus operationalizes the behavioural modeling from Chapter 7, ensuring economic consistency across ALM components.

9.8 FTP and Interest Rate Risk (IRRBB)

FTP interacts directly with **IRRBB metrics**.

By transferring a fixed-term BTR to lending and deposit businesses, the ALM unit isolates the **interest rate position** centrally in Treasury, which can then be **hedged through derivatives** or **structural positioning**.

This structure:

- Prevents uncontrolled mismatches at business-unit level;
- Concentrates rate risk where expertise and instruments exist;
- Simplifies hedge accounting (FVH or CFH frameworks).

In effect, **FTP is the operational backbone of IRRBB management**.

9.9 FTP and Liquidity Risk

FTP also embeds **liquidity incentives**:

- Business units funding long-term loans with short-term sources incur liquidity charges;
- Units raising stable deposits receive liquidity credits;
- Treasury acts as a *liquidity clearing house*, balancing internal flows and ensuring regulatory ratio compliance.

This creates an **internal market for liquidity**, promoting optimal maturity transformation within risk appetite limits.

9.10 FTP and Performance Measurement

FTP-adjusted profitability forms the basis for **risk-adjusted performance metrics** such as:

$$\text{RAROC} = \frac{\text{Net Income after FTP}}{\text{Economic Capital}}$$

$$\text{RORWA} = \frac{\text{Net Income after FTP}}{\text{Risk-Weighted Assets}}$$

This allows consistent comparison of business lines, factoring both funding costs and risk consumption.

ALM thus becomes an enabler of **value-based management**, aligning capital allocation with strategic objectives.

9.11 FTP Governance

An effective FTP framework requires:

- **Clear governance** through ALCO and Treasury policies;
- **Transparent methodologies** shared with business units;
- **Regular recalibration** of curves and liquidity premiums;
- **Audit trail** and documentation for supervisory review (EBA/GL/2018/02).

Roles are typically divided as follows:

Function	Responsibility
ALM/Treasury	Curve definition, calibration, aggregation
Risk Management	Model validation, liquidity assumptions
Finance/Controlling	Performance reporting, profitability analysis
Business Lines	Product pricing, behavioural input
Internal Audit	Independent review

9.12 FTP in Practice: Example

Product	Client Rate (%)	BTR (%)	LP (bps)	RA (bps)	FTP Rate (%)	Margin (%)
Retail mortgage (10Y)	4.20	3.50	+0.30	+0.10	3.90	+0.30
SME loan (5Y)	4.50	3.40	+0.20	+0.05	3.65	+0.85
Retail deposit (3Y)	1.80	2.20	−0.10	0	2.10	−0.30

This table illustrates how FTP quantifies the *economic contribution* of each product, isolating true performance from structural funding effects.

9.13 Challenges and Evolving Practices

Modern FTP frameworks face new challenges:

- **Low/negative rate environments:** distortion of term structures and optionalities;
- **Digital transformation:** rapid deposit flows altering behavioural durations;
- **Regulatory scrutiny:** supervisors requiring transparency and consistency;
- **ESG integration:** incorporating sustainability premiums into funding spreads.

Advanced institutions are adopting **dynamic FTP models** that link transfer rates to market and liquidity indicators in real time.

9.14 Strategic Role of FTP in ALM

FTP transforms ALM from a compliance function into a **strategic steering mechanism**. Through it, the bank can:

- Optimize **risk-adjusted profitability** across portfolios;
- Align product pricing with **liquidity and rate risk constraints**;
- Incentivize **stable funding behaviours**;
- Support **capital efficiency** and **sustainable growth**.

In this sense, FTP is not merely a pricing tool—it is the **internal language of risk, liquidity, and value** within the bank.

9.15 References

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Chapter 10 – Capital Management, ICAAP, and SREP Framework

10.1 Introduction

Capital management represents the **ultimate layer of financial resilience and strategic steering** in a modern bank's balance sheet architecture.

While liquidity ensures short-term survival and ALM governs structural risks, **capital absorbs unexpected losses**, sustaining the bank's solvency, market confidence, and regulatory compliance.

Under the Basel III and Basel IV frameworks, and as refined by the **EBA's Supervisory Review and Evaluation Process (SREP)**, banks are required to maintain sufficient capital to cover both **Pillar 1** (quantitative minimums) and **Pillar 2** (bank-specific) risks.

To ensure the adequacy of these buffers, institutions must develop a comprehensive **Internal Capital Adequacy Assessment Process (ICAAP)**, aligned with their risk appetite, strategy, and macro-financial environment.

Capital management thus integrates the **prudential, economic, and strategic** dimensions of bank management, bridging the regulatory view of solvency with the internal view of value creation.

10.2 Objectives of Capital Management

The objectives of an effective capital management framework include:

1. **Ensuring solvency** through the maintenance of adequate capital buffers relative to risks;
2. **Optimizing capital efficiency**, maximizing shareholder value within regulatory constraints;
3. **Supporting strategic growth**, by allocating capital to profitable and sustainable business lines;
4. **Maintaining market confidence**, through transparent reporting and stable capital ratios;
5. **Aligning with supervisory expectations**, as formalized in the SREP and the EBA's guidelines on ICAAP.

In this sense, capital management serves as both a *compliance function* and a *strategic management tool*.

10.3 Regulatory Framework: Basel and SREP

The regulatory capital framework is structured along three complementary pillars:

Pillar	Focus	Regulatory Reference
Pillar 1	Minimum quantitative capital requirements (credit, market, operational risk)	Basel III / CRR2
Pillar 2	Internal capital adequacy assessment (ICAAP) and supervisory evaluation (SREP)	EBA/GL/2022/11
Pillar 3	Disclosure and market discipline	CRR2 Part Eight

The **SREP** (Supervisory Review and Evaluation Process) ensures that each institution's capital and liquidity positions are **commensurate with its risk profile**.

Supervisors assess the ICAAP and ILAAP, determine the **Pillar 2 Requirement (P2R)** and **Pillar 2 Guidance (P2G)**, and may impose qualitative measures where deficiencies are observed.

10.4 The Internal Capital Adequacy Assessment Process (ICAAP)

The ICAAP is the **core tool of internal capital management**, integrating risk measurement, capital planning, and governance.

According to the EBA Guidelines (EBA/GL/2022/11), the ICAAP must be:

- **Forward-looking**, covering at least a three-year horizon;
- **Comprehensive**, encompassing all material risks (including interest rate, liquidity, concentration, and business risk);
- **Proportionate**, reflecting the size, complexity, and systemic relevance of the bank;
- **Integrated**, linking strategic planning, budgeting, and risk appetite.

The **ICAAP cycle** can be summarized as follows:

1. **Identification and quantification of material risks;**
2. **Assessment of internal capital needs** under baseline and stressed scenarios;
3. **Aggregation of capital requirements** and comparison with available capital;
4. **Capital planning**, including dividends, issuances, and contingency measures;
5. **Governance and approval** by the Board and ALCO.

10.5 Economic Capital and Risk Coverage

While regulatory capital (Basel III) is based on standardized or model-based formulas, **economic capital** represents the *bank's internal assessment of capital required to cover unexpected losses at a chosen confidence level*.

Economic capital is typically estimated for each risk type:

Risk Type	Measurement Method	Confidence Level (Typical)
Credit Risk	Credit VaR, unexpected loss	99.9%
Market Risk	Historical or Monte Carlo VaR	99%
IRRBB	Δ EVE sensitivity or economic value-at-risk	99%
Operational Risk	Loss distribution approach	99.9%
Liquidity Risk	Stress-based shortfall models	99%
Business / Strategic Risk	Scenario-based qualitative methods	95–99%

The total economic capital is the **aggregated amount required to maintain solvency with a probability exceeding 99.9% over one year**, adjusted for diversification effects.

10.6 Capital Planning and Projections

Capital planning links **strategic objectives** with **capital availability and constraints**. The process involves forecasting:

- **Risk-weighted assets (RWA)** evolution;
- **Profitability (NII, trading, fees)**;
- **Dividend and distribution policies**;
- **Funding and leverage metrics**;
- **Stress scenarios** (macro and idiosyncratic).

Example: Capital Forecast Summary (3-Year Horizon)

Metric	2024	2025 (Base)	2026 (Stress)
CET1 Ratio	14.3%	13.5%	11.8%
Leverage Ratio	5.2%	5.0%	4.7%
Total Capital Ratio	18.1%	17.3%	15.5%
RWA (EUR bn)	120	126	132
Profit (EUR mn)	1,450	1,380	950

The ICAAP must demonstrate that even under **severe but plausible** stress, capital ratios remain above regulatory minima and internal thresholds.

10.7 Integration with Risk Appetite and ALM

Capital adequacy cannot be assessed in isolation—it must be **consistent with the bank’s risk appetite and balance sheet structure**. ALM provides the link between **economic risk measures** (ΔEVE , ΔNII) and **capital requirements**.

For instance:

- **Interest rate risk (IRRBB)** impacts both net interest income and the economic value of equity;
- **Liquidity stress** affects funding spreads and solvency under market pressure;
- **Credit deterioration** impacts capital via higher risk weights and provisions.

Through ICAAP integration, ALM indicators translate into **capital consumption metrics**, enabling proactive management.

10.8 ICAAP and SREP Interaction

The **SREP** assesses both the *quantitative adequacy* and the *qualitative robustness* of the ICAAP. Supervisors evaluate:

1. The governance structure and Board involvement;
2. The methodological soundness of internal models;
3. The coherence between ICAAP outcomes and strategic planning;
4. The credibility of stress testing and recovery plans.

The result is a **supervisory decision** assigning:

- **Pillar 2 Requirement (P2R)** — binding additional capital;
- **Pillar 2 Guidance (P2G)** — non-binding buffer for stress resilience;
- **Qualitative measures** (e.g., governance improvements, model reviews).

10.9 Stress Testing in ICAAP

Stress testing is a **core analytical pillar** of ICAAP, designed to assess the impact of adverse macroeconomic scenarios on capital adequacy.

Typical stress scenarios include:

- Sudden interest rate shocks (parallel and non-parallel);
- GDP contraction and credit spread widening;
- Liquidity dry-up or deposit outflows;
- Asset price shocks in sovereign or corporate exposures.

Example – IRRBB Stress Scenario

Shock Type	Δ EVE / Tier 1	Δ NII / Tier 1 (12m)	Capital Impact
Parallel +200 bps	–12.5%	–4.3%	–80 bps CET1
Parallel –200 bps	+8.9%	+3.1%	+55 bps CET1
Steeper	–6.4%	–2.7%	–40 bps CET1

This linkage demonstrates how **ALM sensitivities feed directly into capital projections**, closing the loop between structural risk and solvency.

10.10 Governance and Documentation

EBA (EBA/GL/2022/11) emphasizes the need for a **robust governance framework**, covering:

- Clear **roles and responsibilities** (Board, Risk Committee, ALCO, CFO);
- **Formal documentation** of methodologies and assumptions;
- **Independent validation** by Internal Audit and Risk Control;
- **Regular back-testing** of projections and stress outcomes;
- **Comprehensive reporting** to management and supervisors.

The **Board of Directors** bears ultimate accountability for ICAAP adequacy and must approve the annual ICAAP report.

10.11 Capital Allocation and RAROC Integration

Capital allocation connects ICAAP outcomes to **business strategy and performance measurement**.

Using **Risk-Adjusted Return on Capital (RAROC)** or **Return on Risk-Weighted Assets (RORWA)** metrics, the bank ensures that scarce capital is deployed toward **value-generating activities**.

$$RAROC = \frac{\text{Risk-Adjusted Net Income}}{\text{Economic Capital}}$$

$$RORWA = \frac{\text{Net Income}}{\text{Risk-Weighted Assets}}$$

The **ALCO** and **Strategic Planning Committee** use these metrics to:

- Identify underperforming portfolios;
- Reprice or reduce high-risk exposures;
- Support capital-efficient product design;
- Optimize business mix under regulatory constraints.

10.12 Linkage Between ICAAP and ILAAP

The **Internal Liquidity Adequacy Assessment Process (ILAAP)** complements ICAAP by ensuring that **liquidity and funding** risks are covered by adequate resources and governance. Both processes share:

- Common scenarios and stress assumptions;
- Joint governance under the ALCO;
- Consistent communication to the supervisor.

An integrated **ICAAP–ILAAP** view allows holistic management of **solvency and liquidity under stress**, which is central to EBA’s supervisory philosophy.

10.13 Emerging Trends in Capital Management

Modern capital management must adapt to several evolving dynamics:

- **Basel IV implementation**, introducing output floors and revised credit risk models;
- **Climate and ESG risks**, increasingly incorporated into capital planning (ECB Guide 2022);
- **Macroprudential buffers** (CCyB, O-SII, SyRB) requiring forward-looking allocation;
- **Digitalization of stress testing** using machine learning for scenario design;
- **Integration of ALM and ICAAP data architectures**, enabling near-real-time capital forecasting.

10.14 Conclusion

Capital management, through ICAAP and its supervisory evaluation under SREP, represents the **culmination of the ALM process**.

It transforms balance sheet analytics—interest rate, liquidity, and funding—into strategic decisions on **risk appetite, capital allocation, and value creation**.

In a world of rising macro volatility, regulatory scrutiny, and digital transformation, the integration of **ALM, ICAAP, and capital strategy** is essential not only for compliance, but for **sustainable banking resilience**.

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Chapter 11 – Liquidity Risk Management and ILAAP

11.1 Introduction

Liquidity is the lifeblood of banking. Unlike solvency, which concerns the long-term ability to absorb losses, **liquidity risk** refers to a bank's short-term ability to meet its financial obligations as they come due, without incurring unacceptable losses or triggering a loss of market confidence. The **Global Financial Crisis (GFC) of 2007–2009** demonstrated that even solvent institutions can fail rapidly when liquidity dries up.

In the post-crisis regulatory framework, liquidity management has evolved from a purely treasury-driven operational concern to a central element of the **prudential supervision architecture**.

The **Liquidity Coverage Ratio (LCR)** and **Net Stable Funding Ratio (NSFR)** introduced under the **Basel III framework (BCBS, 2013)**, together with the **Internal Liquidity Adequacy Assessment Process (ILAAP)** mandated by the **European Banking Authority (EBA, 2018)**, form the cornerstone of the modern regulatory liquidity regime.

11.2 The Nature of Liquidity Risk

Liquidity risk arises from the **maturity transformation function** of banks: borrowing short-term (typically via deposits and market funding) and lending or investing long-term.

It can be classified into two primary categories:

1. **Funding Liquidity Risk** – The inability to meet payment obligations due to lack of funding sources (e.g., deposit outflows, maturing wholesale debt, margin calls).
2. **Market Liquidity Risk** – The inability to sell assets at or near their fair value because of market illiquidity, widening bid-ask spreads, or limited depth in secondary markets.

The interaction between these two forms can create **liquidity spirals** (Brunnermeier & Pedersen, 2009), where funding stress forces asset sales, further depressing market prices and increasing collateral haircuts.

11.3 Internal Liquidity Adequacy Assessment Process (ILAAP)

The **ILAAP** is a key component of the **Supervisory Review and Evaluation Process (SREP)**, introduced by the EBA to complement ICAAP (Internal Capital Adequacy Assessment Process). It requires banks to **self-assess** their liquidity risk management frameworks, ensuring that they

maintain adequate liquidity buffers and funding structures aligned with their risk profile and business model.

The **EBA Guidelines on ILAAP (EBA/GL/2016/10 and 2021 revision)** identify four core principles:

1. **Comprehensiveness** – Coverage of all liquidity and funding risks, including intraday liquidity, contingent risks, and off-balance-sheet commitments.
2. **Proportionality** – Tailoring the framework to the bank's size, complexity, and systemic relevance.
3. **Forward-looking orientation** – Incorporation of stress testing and scenario analysis to anticipate liquidity stress events.
4. **Integration with governance and strategic planning** – Alignment with the bank's business model, risk appetite, and ICAAP.

The ILAAP must be documented in a formal **ILAAP Report**, reviewed annually by senior management and the Board, and submitted to the supervisory authority (ECB or national competent authority) as part of the SREP package.

11.4 Components of a Robust Liquidity Risk Framework

A sound **liquidity risk management framework** integrates both **quantitative** and **qualitative** elements, typically structured as follows:

11.4.1 Governance

- Board and ALCO (Asset & Liability Committee) oversight.
- Clearly defined liquidity risk appetite and tolerance thresholds.
- Integration with capital planning and recovery/resolution frameworks.

11.4.2 Measurement

- Daily monitoring of LCR, NSFR, and internal liquidity ratios.
- Cash flow projections by currency, entity, and maturity bucket.
- Stress testing under idiosyncratic and systemic shocks.

11.4.3 Monitoring Tools

- Concentration limits on funding sources and counterparties.
- Collateral eligibility and encumbrance analysis.
- Early warning indicators (EWIs) for deposit outflows or market stress.

11.4.4 Contingency Funding Plan (CFP)

- Escalation procedures and communication plans.
- Identification of available funding actions (e.g., repo, central bank facilities).
- Definition of liquidity triggers for activation.

11.5 Stress Testing and Scenario Design

Liquidity stress testing under ILAAP must incorporate both **idiosyncratic shocks** (e.g., rating downgrade, reputational event) and **systemic stress scenarios** (e.g., interbank market freeze, sovereign crisis).

Key elements include:

- **Time horizon:** typically 30 days (LCR) and 90 days for extended scenarios.
- **Granularity:** projection of contractual and behavioral cash flows.
- **Assumptions:** run-off rates, market haircuts, and central bank access.
- **Reverse stress tests:** identifying conditions under which the bank would fail liquidity requirements.

Results must feed into the **liquidity buffer calibration** and **funding strategy**, ensuring that management actions are realistic and timely.

11.6 Interaction between ILAAP and ICAAP

While ICAAP focuses on **capital adequacy** and solvency under stress, ILAAP addresses **liquidity and funding adequacy**.

The two processes are complementary and increasingly integrated under supervisory expectations (ECB, *Guide to ICAAP and ILAAP*, 2020):

Dimension	ICAAP	ILAAP
Objective	Ensure capital adequacy	Ensure liquidity adequacy
Risk focus	Credit, market, operational	Liquidity, funding, concentration
Time horizon	Multi-year (3–5 years)	Short-to-medium term (1–2 years)
Output	Capital plan, risk appetite	Funding plan, liquidity buffer
Stress tests	Capital depletion	Liquidity outflows and funding gaps

Integration ensures a **holistic balance sheet view**, where capital and liquidity interact dynamically. For instance, **capital erosion** reduces market confidence and funding access, while **liquidity stress** may force asset fire sales impacting capital ratios.

11.7 The Role of the ALCO in Liquidity Management

The **Asset & Liability Committee (ALCO)** acts as the central governance body for balance sheet and liquidity management.

Its responsibilities include:

- Reviewing LCR/NSFR and internal liquidity ratios.
- Approving funding and investment plans.
- Ensuring consistency with risk appetite and regulatory expectations.
- Coordinating ICAAP/ILAAP integration.

ALCO's forward-looking decisions on duration, funding mix, and buffer composition directly shape the bank's liquidity resilience and profitability trade-offs.

11.8 Key Supervisory Expectations and Best Practices

According to the **EBA SREP methodology** and **ECB guidance**, best practices for liquidity management include:

- Maintaining an **adequate buffer** of HQLA calibrated to stress scenarios.
- Ensuring **diversified and stable funding**, limiting reliance on volatile wholesale markets.
- Monitoring **intraday liquidity** positions and collateral encumbrance.
- Conducting **regular independent validation** of liquidity models and assumptions.
- Establishing a **liquidity transfer pricing (LTP)** mechanism to allocate liquidity costs across business lines.

Supervisors increasingly assess banks' **liquidity governance maturity** using qualitative scores, considering the consistency of the ILAAP with the institution's actual risk profile and management practices.

11.9 Emerging Topics: Climate and ESG Liquidity Risks

Recent regulatory developments (EBA Discussion Paper, 2023) highlight the potential liquidity implications of **climate-related and ESG risks**:

- Physical climate risks can impact asset valuations and collateral liquidity.
- Transition risks can affect funding stability in carbon-intensive sectors.
- Investors' and depositors' preferences may shift rapidly, creating new **reputational liquidity shocks**.

Banks are therefore expected to **integrate ESG considerations** into liquidity stress testing and ILAAP governance, particularly in the context of **climate scenario analysis** (ECB, *Guide on Climate-Related Risks*, 2021).

11.10 Conclusion

Liquidity risk management has evolved from a reactive treasury function into a **strategic and supervisory pillar** of bank resilience.

The ILAAP provides a structured, forward-looking framework to ensure that liquidity and funding risks are identified, measured, and managed consistently with the institution's risk appetite and business strategy.

Its integration with ICAAP allows banks to achieve a **comprehensive view of balance sheet adequacy**, bridging solvency and liquidity in a coherent governance system.

As financial markets evolve—particularly with digital transformation, ESG pressures, and the normalization of monetary policy—the **strategic management of liquidity** will remain central to the stability and competitiveness of modern banks.

11.11 References

- Basel Committee on Banking Supervision (BCBS) (2013), *Basel III: The Liquidity Coverage Ratio and Liquidity Risk Monitoring Tools*.
- European Banking Authority (EBA) (2016, 2021), *Guidelines on ILAAP under the SREP Framework*.
- European Central Bank (ECB) (2020), *Guide to the Internal Liquidity Adequacy Assessment Process (ILAAP)*.
- Brunnermeier, M., & Pedersen, L. (2009), *Market Liquidity and Funding Liquidity*, *Review of Financial Studies*, 22(6), 2201–2238.

- EBA (2023), *Discussion Paper on the Role of Environmental Risks in the Prudential Framework*.

Chapter 12 – Integration of ICAAP and ILAAP: The Holistic Balance Sheet View

12.1 Introduction

The integration of **capital** and **liquidity adequacy frameworks** — the **Internal Capital Adequacy Assessment Process (ICAAP)** and the **Internal Liquidity Adequacy Assessment Process (ILAAP)** — represents one of the most critical developments in modern bank risk governance.

While these two processes were originally conceived as distinct supervisory requirements under **Pillar II of Basel III**, recent regulatory evolution and market experience have shown that **capital and liquidity risks are interdependent** and must be analyzed within a **unified balance sheet management framework**.

The **European Central Bank (ECB, 2020)** explicitly encourages banks to move toward **holistic balance sheet management**, in which ICAAP and ILAAP jointly form the foundation of the institution's **strategic planning, risk appetite, and stress testing**.

This chapter explores how banks can integrate the two frameworks effectively to ensure **resilience, consistency, and strategic coherence** across solvency and liquidity dimensions.

12.2 Conceptual Interdependence between Capital and Liquidity

Although ICAAP and ILAAP serve distinct prudential purposes, their **interdependencies** are numerous and significant:

Dimension	ICAAP Focus	ILAAP Focus	Interconnection
Objective	Capital adequacy under stress	Liquidity and funding adequacy	Liquidity shocks can erode capital; capital depletion weakens funding access
Primary Risk Drivers	Credit, market, operational, concentration	Funding, refinancing, market liquidity	Market stress transmits across both
Time Horizon	Medium- to long-term (3–5 years)	Short- to medium-term (1–2 years)	Strategic capital plan influences funding plan
Management Tool	Capital plan, ICAAP report	Funding and liquidity plan, ILAAP report	ALM acts as the link between the two
Supervisory Outcome	Pillar II capital add-on	Liquidity guidance and qualitative measures	Both influence SREP scores and capital/liquidity buffers

The interplay between solvency and liquidity becomes evident during **stress conditions**:

- A **liquidity shortage** may force asset sales at depressed prices, leading to **capital losses**.
- Conversely, **capital deterioration** undermines market confidence, restricting access to funding and triggering liquidity stress.

These dynamics reinforce the need for a **joint assessment and governance approach**.

12.3 Regulatory and Supervisory Framework

The **Basel Committee on Banking Supervision (BCBS, 2014)** and the **EBA (2021)** both stress that ICAAP and ILAAP should be “**two pillars of a single institution-wide framework**” for ensuring sound internal management.

Key regulatory documents shaping this integration include:

- **ECB (2020)** – *Guide to the Internal Capital and Liquidity Adequacy Assessment Processes (ICAAP and ILAAP)*
- **EBA SREP Guidelines (2018)** – *Supervisory Review and Evaluation Process Framework*
- **BCBS (2019)** – *Principles for the Sound Management of Liquidity Risk and Capital Planning*

Supervisors expect banks to demonstrate **cross-consistency** in their ICAAP and ILAAP documentation, particularly regarding:

- Scenario design and macroeconomic assumptions
- Management actions and contingency measures
- Alignment of stress testing horizons and methodologies
- Internal governance, data quality, and model validation

The **Supervisory Review and Evaluation Process (SREP)** integrates findings from both ICAAP and ILAAP to determine a bank’s **overall SREP score**, which influences **Pillar II capital additions** and **liquidity guidance (P2G)**.

12.4 Building an Integrated ICAAP–ILAAP Framework

An effective integration approach typically involves three main stages:

(1) Strategic Alignment

Both processes must stem from the bank's **business model** and **risk appetite framework (RAF)**.

This requires:

- Common macroeconomic and financial assumptions (e.g., interest rate paths, GDP growth, market volatility).
- Consistency between **capital planning** and **funding plans** over the strategic horizon.
- Incorporation of **ALCO decisions** into both ICAAP and ILAAP cycles.

(2) Methodological Consistency

The methodologies applied in the two frameworks must be internally coherent:

- **Stress testing** scenarios must use consistent drivers (e.g., rate shocks, credit spreads, deposit outflows).
- **Data architecture** must ensure shared sources and traceability of assumptions.
- **Risk aggregation** should allow mapping liquidity effects (e.g., funding cost increases) onto capital impacts (e.g., lower NII, MTM losses).

(3) Governance Integration

The **Board of Directors** and the **Risk Committee** should oversee both ICAAP and ILAAP jointly, ensuring:

- A single **reporting cycle** for both documents.
- Coherence between the **Risk Appetite Statement (RAS)** and **Capital/Liquidity Buffers**.
- Centralized ALM coordination and treasury–risk alignment.

This ensures that ICAAP and ILAAP are not siloed exercises but **mutually reinforcing management tools**.

12.5 Role of Stress Testing in Integration

Stress testing is the primary operational bridge between ICAAP and ILAAP. Under the **ECB's integrated stress testing expectations (ECB, 2021)**, banks must design scenarios that simultaneously capture:

- **Capital impacts** (e.g., credit losses, MTM valuation effects, RWA increases).
- **Liquidity impacts** (e.g., deposit withdrawals, market funding spread widening).

A combined stress test must illustrate how **liquidity shocks affect solvency**, and vice versa. For instance:

- In a **systemic stress scenario**, a widening of sovereign spreads can both increase RWA (capital effect) and reduce collateral value (liquidity effect).
- A **funding market freeze** can lead to higher reliance on central bank liquidity, increasing leverage and potentially lowering capital ratios.

These feedback loops require **dynamic balance sheet modeling**—often embedded within the bank’s **ALM system**—to capture the time-phased evolution of both capital and liquidity metrics.

12.6 Integration via ALM: The Balance Sheet Nexus

Asset & Liability Management (ALM) acts as the operational platform for ICAAP–ILAAP integration.

Through ALM, a bank links **earnings, capital, and liquidity** via consistent modeling of balance sheet dynamics.

Key mechanisms include:

- **Interest rate risk in the banking book (IRRBB)**: affects both NII (income perspective) and EVE (economic value perspective), bridging profitability and capital.
- **Funding strategy decisions**: determine liquidity buffer needs and impact funding costs, which in turn affect NII and capital generation.
- **Capital and liquidity stress tests**: use ALM data to simulate behavioral responses of assets and liabilities under macro stress.

Hence, ALM functions as both a **risk quantification engine** and a **strategic coordination mechanism**, ensuring coherence between ICAAP and ILAAP assumptions.

12.7 The Holistic Balance Sheet View

The **holistic view** integrates capital, liquidity, and interest rate risk management into a **single economic model of balance sheet resilience**.

Conceptually, it links three key dimensions:

Dimension	Indicator	Strategic Focus
Capital Adequacy	CET1 Ratio, Total Capital Ratio	Solvency, RWA efficiency
Liquidity Adequacy	LCR, NSFR, Internal Liquidity Ratios	Short-term resilience, funding stability
Interest Rate Risk (IRRBB)	Δ NII, Δ EVE	Earnings stability, economic value preservation

The holistic balance sheet framework allows management and supervisors to evaluate the **trade-offs** between these dimensions.

For example:

- A **larger liquidity buffer** improves LCR but may reduce profitability.
- Extending **asset duration** increases NII sensitivity and EVE volatility, affecting both capital and liquidity usage.
- Increasing **capital buffers** enhances solvency but may limit lending and liquidity creation.

Thus, the **integrated ICAAP–ILAAP model** provides a coherent decision-making framework for balancing resilience and profitability.

12.8 Supervisory Expectations for Integration

The **ECB’s 2020 ICAAP/ILAAP Guide** outlines explicit expectations for integration, emphasizing:

1. **Consistency in management actions** across capital and liquidity domains.
2. **Common scenario frameworks** for macroeconomic, market, and idiosyncratic stresses.
3. **Shared governance** via ALCO and the Risk Committee.
4. **Cross-validation** between ICAAP and ILAAP assumptions (e.g., dividend policy, funding costs, recovery triggers).

In recent **SREP cycles**, supervisors have begun explicitly assessing the **degree of integration** between the two frameworks, assigning higher qualitative scores to banks demonstrating joint stress testing and management reporting.

12.9 Integration Challenges

Despite regulatory encouragement, many banks face practical challenges in achieving full integration:

- **Data fragmentation** between Treasury, Risk, and Finance functions.
- **Different time horizons** and modeling techniques (static vs dynamic).
- **Inconsistent governance cycles** and documentation timelines.
- **Complexity of behavioral modeling**, particularly for non-maturing deposits and contingent liquidity.

Leading institutions have responded by developing **integrated capital and liquidity dashboards**, **ALM-driven simulation engines**, and **joint ICAAP–ILAAP Committees** to ensure coordination and model transparency.

12.10 Future Developments: Toward an Integrated Resilience Framework

The next evolution in prudential supervision is the convergence of ICAAP and ILAAP into a **single “Internal Resilience Assessment Framework” (IRAF)**, as discussed in ECB technical workshops (2023–2024).

This would formally integrate solvency, liquidity, and business model risk assessments within one strategic process, with:

- Unified scenario architecture.
- Joint reporting to the Board and supervisors.
- Quantitative resilience indicators combining capital, liquidity, and profitability metrics.

This vision underscores the shift from regulatory compliance toward **true strategic risk integration** across the entire balance sheet.

12.11 Conclusion

The integration of ICAAP and ILAAP represents the culmination of modern bank risk governance — moving from siloed risk management toward a **unified, forward-looking, and strategic framework**.

Through ALM and consistent stress testing, banks can achieve a **holistic view of resilience**, balancing capital strength, liquidity adequacy, and earnings stability.

Such integration not only satisfies supervisory expectations but also enhances **strategic agility**, allowing institutions to anticipate shocks, optimize capital and funding resources, and maintain stakeholder confidence.

In an environment of renewed interest rate volatility, ESG transition pressures, and digital transformation, a **holistic balance sheet view** is no longer optional — it is the **core of sustainable banking management**.

12.12 References

- European Central Bank (2020), *Guide to the Internal Capital and Liquidity Adequacy Assessment Processes (ICAAP and ILAAP)*.
- European Banking Authority (2018, 2021), *SREP Guidelines and ICAAP/ILAAP Integration Expectations*.
- Basel Committee on Banking Supervision (2019), *Principles for Sound Liquidity and Capital Risk Management*.
- Brunnermeier, M. & Pedersen, L. (2009), *Market Liquidity and Funding Liquidity*, *Review of Financial Studies*, 22(6), 2201–2238.
- ECB Technical Workshop (2023), *Toward a Unified Internal Resilience Framework (IRAF)*.

Part II

Applications and Case Studies

Chapter 13 – Case Study 1: Interest Rate Risk and ALM Optimization under Monetary Tightening

13.1 Introduction

The post-pandemic period marked the fastest global monetary tightening cycle in decades. Central banks across advanced economies—most notably the **Federal Reserve**, the **European Central Bank (ECB)**, and the **Bank of England**—raised policy rates by several hundred basis points between 2022 and 2023 in response to persistent inflationary pressures.

For banks, this new regime fundamentally reshaped **balance sheet dynamics**. Net interest margins initially expanded due to asset repricing, but **interest rate risk** and **mark-to-market volatility** increased sharply, especially for institutions holding long-duration sovereign or mortgage portfolios.

This case study examines how a representative mid-size **European commercial bank (“Bank A”)** optimized its ALM positioning during the tightening phase of 2022–2023, balancing profitability, duration risk, and capital impacts under regulatory constraints (IRRBB and ICAAP/ILAAP integration).

13.2 Macro and Market Context

Variable	Q1 2022	Q2 2022	Q3 2022	Q4 2022	Q1 2023
ECB Deposit Facility Rate	–0.50%	0.00%	1.25%	2.00%	3.00%
2Y EUR Swap Rate	–0.10%	1.00%	2.20%	2.90%	3.30%
10Y EUR Swap Rate	0.30%	1.60%	2.60%	2.85%	3.05%
Inflation (YoY, Eurozone)	5.8%	8.1%	9.9%	8.4%	6.9%
GDP Growth (YoY)	4.0%	3.5%	2.2%	1.4%	0.8%
10Y Bund Yield	0.30%	1.50%	1.90%	2.10%	2.30%

Market interpretation:

- **Curves flattened**, signaling expectations of economic slowdown.
- **Inflation persistence** forced central banks to accelerate tightening.
- Bond prices fell sharply, impacting FVOCI portfolios.
- Funding costs rose, especially for wholesale funding.

13.3 Starting ALM Position of Bank A

Balance Sheet Item	Amount (€bn)
Loans to customers (fixed rate)	12.0
Loans to customers (variable rate)	8.0
Securities portfolio (FVOCI + HTC)	5.0
Interbank assets	2.0
Customer deposits (non-maturing)	15.0
Term deposits	4.0
Wholesale funding	3.0
Equity	2.0

Key ALM indicators at Q1 2022:

- **Duration Gap:** +0.6 years
- **ΔEVE / Tier 1:** –11% (sensitivity to –200 bp shock)
- **ΔNII / Tier 1:** +4.8% (benefit from rate hikes)
- **Liquidity Coverage Ratio (LCR):** 145%
- **NSFR:** 112%

Interpretation:

The bank's balance sheet was moderately **asset-sensitive**, benefiting from higher short-term rates through NII expansion, but exposed to **EVE losses** in a downward rate shock.

13.4 Strategic Challenge

As rates rose rapidly, **mark-to-market losses** accumulated on the FVOCI portfolio, eroding **OCI reserves** and threatening the **CET1 ratio**.

Simultaneously, **customer deposit repricing** lagged, boosting margins but raising concerns about potential outflows as competition for deposits intensified.

The **ALCO** faced a dual objective:

1. Protect capital and limit ΔEVE volatility.
2. Maintain or enhance NII to sustain earnings in a high-rate environment.

13.5 Management Discussion and Strategic Options

The **ALM Committee (ALCO)** considered two alternative strategies:

Strategy	Description	Pros	Cons
A – Defensive (Duration Reduction)	Shorten asset duration by selling long-term bonds (FVOCI) and re-investing in short-dated government securities or floating-rate notes (FRNs).	Reduces Δ EVE sensitivity; improves LCR; fast impact on capital.	Lowers carry income; reinvestment risk if rates fall again.
B – Opportunistic (Lock-in Carry)	Increase fixed-rate exposure by buying 5–7Y BTPs and extending duration moderately.	Locks higher yields; supports medium-term NII; beneficial if rates stabilize.	Increases Δ EVE sensitivity; higher MTM volatility; liquidity absorption.

13.6 Quantitative Simulation

Scenario	Duration Gap (yrs)	Δ EVE/Tier 1	Δ NII/Tier 1 (12m)	LCR	Expected CET1 Impact
Initial (Q1 2022)	+0.6	–11%	+4.8%	145%	CET1 13.2%
Strategy A (shorter)	+0.3	–6.0%	+2.9%	160%	CET1 +0.2pp
Strategy B (longer)	+0.9	–15.0%	+6.1%	138%	CET1 –0.3pp

Interpretation:

- **Strategy A** aligns with prudential risk reduction — mitigates EVE loss and improves liquidity.
- **Strategy B** enhances profitability but heightens capital volatility.

13.7 ALM Committee Decision

After stress testing under both **parallel** and **steepening** scenarios:

- The **ALCO** opted for a **hybrid approach**: partial reduction of FVOCI exposure and simultaneous reinvestment in medium-term sovereigns (3–4Y).
- Complementary **interest rate swaps (payer IRS)** were executed to neutralize excessive duration.
- The net effect was a **duration gap of +0.45 years**, stabilizing both Δ EVE and Δ NII metrics.

This balanced positioning allowed the bank to protect solvency while still capturing part of the yield advantage.

13.8 Operational Execution

1. Portfolio Adjustment:

- Sold €0.8bn long BTPs (10Y) at 2.3% yield; realized –€40m MTM loss.
- Purchased €1.2bn of 3Y FRNs at 3.1% coupon.

2. Derivative Strategy:

- Entered €2.0bn **payer IRS (3Y)** at 2.9% vs Euribor 3M.
- Hedge accounted under **macro fair value hedge (FVH)**.

3. Funding Management:

- Gradual shift of €1bn from wholesale to retail term deposits.
- Enhanced liquidity buffers via repo-eligible assets.

13.9 Results after Two Quarters (Q3 2023)

Indicator	Q3 2023 Result	Change vs Q1 2022
ΔEVE / Tier 1	–8.2%	+2.8pp improvement
ΔNII / Tier 1	+5.5%	+0.7pp increase
CET1 Ratio	13.3%	+0.1pp
LCR	155%	+10pp
Duration Gap	+0.45 years	–0.15 years

Conclusion:

The hybrid strategy proved effective:

- **Risk-return optimization** under volatility;
- **Improved resilience** without compromising NII;
- Alignment with ICAAP/ILAAP consistency principles.

13.10 Lessons Learned

- **Dynamic ALM** is essential: static positions can quickly become obsolete under rapid policy shifts.
- **EVE and NII must be optimized jointly**—reducing one risk dimension may worsen the other.

- **Fair Value Hedge accounting** enables effective economic and accounting alignment.
- **Liquidity implications** must be integrated into ALM simulations: duration changes affect both LCR and NSFR.
- Regulatory communication (ICAAP, ILAAP, SREP) benefits from quantitative consistency in ALM decision-making.

13.11 Concluding Remarks

This case demonstrates how a prudently managed ALM strategy can maintain both profitability and resilience during a monetary tightening cycle.

In a high-volatility environment, balance sheet optimization requires not only tactical adjustments in duration and hedging but also a **strategic integration of capital, liquidity, and accounting frameworks**.

The following chapters will explore other scenarios—from crises and macro shocks to strategic allocation dilemmas—further illustrating how ALM serves as the **core of forward-looking financial management** in modern banking.

Chapter 14 – Case Study 2: Liquidity Stress and Funding Strategy under Market Disruption

14.1 Introduction

Liquidity management is at the heart of banking resilience. Even solvent institutions can face acute liquidity crises if funding markets suddenly dry up or depositors lose confidence. The **2020 COVID crisis**, the **UK gilt turmoil of 2022**, and the **U.S. regional bank failures of 2023** have all reminded the financial community that liquidity risk can emerge abruptly, often as a second-round effect of market volatility or interest rate shocks.

This case study explores how **“Bank B”**, a large European universal bank, responded to a **market-wide liquidity stress** following a sudden repricing of sovereign debt markets. The objective is to illustrate how **funding strategy**, **liquidity buffers**, and **ALM coordination** interact under extreme but plausible scenarios, within the framework of **Liquidity Coverage Ratio (LCR)** and **Net Stable Funding Ratio (NSFR)** regulation.

14.2 Scenario and Market Context

In **Q4 2022**, following a sharp increase in European yields and widening sovereign spreads, funding markets became temporarily dislocated:

Variable	Q3 2022	Q4 2022	Q1 2023
ECB Deposit Facility Rate	1.25%	2.00%	3.00%
3M Euribor	1.35%	2.10%	3.10%
10Y BTP Yield	3.50%	4.80%	4.50%
BTP-Bund Spread	220 bps	260 bps	230 bps
Market Liquidity Index (bid-ask)	1.0	2.8	2.2
Euro IG Credit Spread	95 bps	155 bps	130 bps

Key Features of the Stress Episode:

- Sharp repricing in sovereign and credit markets.
- Repo haircuts on sovereign collateral increased by 30–50 bps.
- Investor demand for long-term bank bonds weakened.

- Retail depositors shifted funds toward money market funds and high-yield term deposits.

14.3 Bank B – Initial Balance Sheet Snapshot (Q3 2022)

Balance Sheet Item	Amount (€bn)
Customer Loans	320
Trading Assets & Securities (HTC + FVOCI)	80
Interbank Assets	40
Total Assets	440
Customer Deposits	260
Wholesale Funding	120
Equity	60
Total Liabilities + Equity	440

Liquidity Indicators (Baseline):

- **LCR:** 138%
- **NSFR:** 115%
- **Liquidity Buffer (HQLA):** €95bn
- **Short-Term Funding (≤1 year):** €70bn
- **Stable Funding (retail deposits + LT debt):** €260bn

14.4 Stress Event: Outflows and Funding Pressure

In late 2022, the bank experienced:

- **Deposit outflows** of €20bn (≈8% of retail base) within three weeks.
- **Haircuts on repo operations** increasing collateral requirements by €10bn.
- **Term debt issuance** temporarily closed due to spread widening.

Liquidity Ratios Under Stress (End-Q4 2022):

- **LCR:** 98%
- **NSFR:** 106%

- **Liquidity Buffer (HQLA):** €78bn
- **Wholesale Roll-over Success Rate:** 65%

The situation called for immediate ALCO intervention to **stabilize liquidity** and **preserve regulatory compliance**.

14.5 ALCO Strategic Options

Strategy	Description	Advantages	Drawbacks / Risks
A – Defensive Liquidity Protection	Increase HQLA by selling FVOCI assets and accessing ECB TLTRO or repo lines.	Improves LCR immediately; strengthens market confidence.	Realizes MTM losses; reduces NII; may signal distress.
B – Funding Diversification	Launch accelerated program of retail term deposits (+€10bn) and issue short-dated covered bonds.	Stabilizes liability structure; supports NSFR; reduces market dependency.	Higher funding costs; delayed impact; execution risk.
C – Balance Sheet Optimization	Securitize part of the loan portfolio (€5bn) and release collateral to free up liquidity.	Frees up encumbered assets; structural liquidity improvement.	Complexity, lead time, regulatory approval required.

14.6 Quantitative Simulation

Scenario	LCR (%)	NSFR (%)	ΔLiquidity Buffer (€bn)	ΔFunding Cost (bps)	ΔNII (12m, €mn)
Baseline	138	115	+0	+0	0
Stress (before actions)	98	106	–17	+45	–320
Strategy A	128	108	+12	+20	–150
Strategy B	122	118	+9	+30	–90
Strategy C	130	120	+14	+25	–60
Combined (A + B)	135	119	+15	+27	–100

Interpretation:

- **Strategy A** quickly restores compliance with LCR but at the expense of profitability.
- **Strategy B** is slower but more sustainable in the medium term.
- **Strategy C** provides structural benefits but requires time.
- The **combined approach (A+B)** offers the best short-term and long-term balance.

14.7 Execution Framework

1. Immediate Liquidity Actions (T+3 Days):

- Activation of **ECB standing facility** for €10bn.
- Sale of €5bn FVOCI sovereigns (average duration 3Y, realized –€120m loss).
- Reallocation of excess reserve balances from subsidiaries.

2. Short-Term Funding Diversification (T+2 Weeks):

- Launch of **€8bn retail term deposit campaign** (+40 bps premium).
- Secured issuance of **€2bn 9-month covered bonds** via repo-eligible structure.

3. Structural Measures (T+2 Months):

- Securitization of €5bn mortgage loans; retained senior tranches as HQLA level 2A.
- Revision of internal **Liquidity Contingency Plan** and stress scenarios.

14.8 Results after Two Quarters (Q2 2023)

Indicator	Before Stress (Q3 2022)	After Actions (Q2 2023)	Change
LCR	138%	142%	+4pp
NSFR	115%	118%	+3pp
HQLA Buffer	€95bn	€110bn	+€15bn
Funding Cost	+45 bps	+25 bps (net)	–20 bps
NII Impact (YoY)**	0	–€110m	Negative but contained
Customer Confidence (measured by deposit inflows)**	Stable	+€3bn new retail inflows	Restored trust

14.9 Lessons Learned

- **Liquidity and interest rate risks are interdependent.** Market shocks can erode both funding access and asset valuations.
- **Regulatory liquidity buffers (LCR, NSFR)** are effective only if dynamically managed; static compliance offers limited protection.
- **Funding diversification**—retail, wholesale, and central bank access—is essential for resilience.

- **Communication strategy** is a key ALM tool: transparency with investors and regulators helps contain contagion risk.
- **Liquidity stress testing** should integrate behavioral deposit dynamics, collateral haircuts, and cross-entity transferability constraints.

14.10 Concluding Remarks

This case study demonstrates how a systemic liquidity shock can challenge even a well-capitalized institution, and how **coordinated ALM governance** can restore stability. The key success factors were **early detection**, **decisive action**, and **balanced trade-offs** between liquidity, profitability, and market confidence.

Ultimately, liquidity management is not only about surviving crises—it is about **designing funding structures that remain stable through all phases of the financial cycle**.

Chapter 15 – Case Study 3: Managing Non-Maturing Deposits (NMDs) and Behavioral Optionality in a Low-Rate Environment

15.1 Introduction

Non-Maturing Deposits (NMDs)—such as sight deposits, savings accounts, and current accounts—represent a **core funding source** for most retail and commercial banks. However, their **undefined contractual maturity** and **rate-insensitive behavior** introduce significant modeling challenges from both an **interest rate risk** and **liquidity risk** perspective.

In a **low or negative interest rate environment**, such as that experienced in the Eurozone between 2015 and 2021, the behavior of NMDs diverges sharply from standard theoretical assumptions: deposit betas decline toward zero, repricing asymmetries intensify, and customer stickiness becomes a strategic buffer for banks' net interest margins.

This case study explores the management of NMDs in a European retail bank (“**Bank C**”) through the lens of **behavioral modeling**, **interest rate sensitivity analysis**, and **replication portfolio construction**, focusing on the trade-offs between **stability**, **profitability**, and **risk measurement consistency** under the **IRRBB** (Interest Rate Risk in the Banking Book) regulatory framework.

15.2 The Strategic Role of NMDs

NMDs are essential to the stability of banks' funding structures because they:

1. Provide **stable and low-cost funding**;
2. Act as a **natural hedge** against long-term fixed-rate assets;
3. Contribute significantly to **liquidity metrics** such as the NSFR and LCR;
4. Exhibit **behavioral duration** that may be longer than their contractual features suggest.

However, these benefits are accompanied by **modeling risks**, particularly under **changing rate regimes**, where deposit outflows, migration toward term deposits, or shifts in customer preferences may emerge abruptly.

15.3 Bank C – Balance Sheet Overview (as of Q4 2020)

Balance Sheet Item	Amount (€bn)
Customer Loans	240
Fixed-Income Securities (HTC + FVOCI)	80
Interbank Assets	15
Total Assets	335
Customer Deposits (NMDs)	220
Term Deposits	20
Wholesale Funding	55
Equity	40
Total Liabilities + Equity	335

Interest Rate Risk Metrics:

- Duration Gap: **+0.35 years**
- ΔEVE (–200 bps): **–€42m**
- ΔNII (–200 bps): **–€18m**
- SOT NII: **–4.2%**

15.4 Behavioral Model for NMDs

Bank C applies a **replication portfolio approach** to translate behavioral features into measurable interest rate sensitivity.

Key Behavioral Assumptions:

Parameter	Description	Value
Core Deposit Share	Stable, non-rate-sensitive balances	80%
Volatile Deposit Share	Rate-sensitive, mobile balances	20%
Average Repricing Beta	Sensitivity to market rate changes	0.15
Average Behavioral Duration	Weighted by core vs. volatile share	3.0 years
Optionality	Early withdrawal & migration elasticity	Medium
Floor Rate	0.00% (cannot fall below zero)	

Behavioral insights:

During the negative rate phase (2019–2021), Bank C observed that deposit volumes remained stable despite declining rates, suggesting strong **relationship stickiness**. However, when market expectations for rising rates appeared in early 2022, **beta asymmetry** reemerged—customers became more responsive to upward rate adjustments than downward ones.

15.5 Replication Portfolio Construction

The ALM desk replicated the NMDs as a **synthetic bond portfolio**:

Maturity Bucket	Weight	Instrument Type
6 months	10%	Floating-rate note (Euribor 3M + 10 bps)
2 years	25%	Fixed-rate government bond
4 years	35%	Fixed-rate government bond
7 years	20%	Fixed-rate covered bond
10 years	10%	Swap receiver position (10Y vs 3M)

The weighted average duration of this portfolio was **3.2 years**, consistent with behavioral analysis.

This portfolio generates **interest rate sensitivity** equivalent to the modeled behavior of deposits while remaining liquid and mark-to-market measurable.

15.6 Stress Test: Rising Rate Environment

Scenario: Parallel +200 bps shock on the yield curve, accompanied by a +25% increase in deposit beta (from 0.15 to 0.19) and a 5% outflow of volatile deposits.

Indicator	Baseline	+200bps Scenario	Change
Customer Deposits (€bn)	220	209	-11
Core/Volatile Mix	80% / 20%	85% / 15%	Core share increases
Behavioral Duration (yrs)	3.0	2.7	-0.3
ΔNII (12m)	-€18m	+€32m	Improves due to rate hike
ΔEVE	-€42m	-€55m	Slight deterioration
SOT NII	-4.2%	-3.1%	Lower risk exposure

Interpretation:

Higher rates increase NII via deposit repricing lag, though long-term EVE sensitivity worsens marginally due to duration shortening. Liquidity position remains stable since outflows are moderate and concentrated in volatile deposits.

15.7 Strategic ALM Actions

Strategic Lever	Objective	Operational Implementation	Expected Effect
Behavioral Model Update	Adjust deposit beta assumptions to reflect asymmetric repricing	Calibrate regression on 2017–2022 data	Improve model realism
Portfolio Hedging	Protect against excessive convexity in replication portfolio	Add pay-fixed IRS 5Y	Reduce EVE volatility
Product Design	Retain core deposits	Introduce loyalty savings accounts (0.10% above floor)	Enhance stability of funding
Liquidity Buffer Optimization	Maintain flexibility amid rate hikes	Increase share of short-term HQLA (T-bills, repo lines)	Strengthen LCR resilience
Governance Enhancement	Align IRRBB and liquidity reporting	Create joint ALCO-IRRBB working group	Integrated risk view

15.8 Quantitative Comparison of Strategies

Scenario	ΔEVE (€/mn)	ΔNII (€/mn)	SOT NII (%)	LCR (%)	NSFR (%)
Baseline	−42	−18	−4.2	136	115
After Behavioral Update	−40	−15	−3.8	137	115
After Hedging	−35	−12	−3.5	135	114
After Product Redesign	−33	−10	−3.0	140	117
Combined Strategy	−30	−8	−2.8	142	118

Result:

The integrated ALM strategy enhances earnings stability and regulatory ratios with manageable trade-offs in profitability.

15.9 Discussion and Insights

1. **Behavioral asymmetry is structural, not cyclical.** Deposit repricing behaves differently in rising versus falling rate environments.
2. **Replication portfolios** remain an effective IRRBB management tool, provided their assumptions are reviewed frequently.
3. **Hedging convexity** through interest rate swaps can mitigate excessive volatility in EVE, especially under curve steepening.
4. **Customer behavior analysis** is becoming a key risk factor—ALCOs must integrate behavioral economics into balance sheet decisions.
5. **Regulatory alignment** (EBA IRRBB Guidelines 2022) requires explicit model governance and independent validation of NMD modeling.

15.10 Conclusion

Effective management of Non-Maturing Deposits combines **quantitative modeling** and **behavioral understanding**.

For banks, NMDs are both an **asset and a liability**—a source of funding stability but also of embedded optionality that must be carefully measured, hedged, and reported.

In a world where monetary regimes shift quickly, **behavioral calibration** and **dynamic ALM governance** are the true hedges against uncertainty.

Chapter 16 – Case Study 4: Duration Gap Management and the Trade-Off Between Δ EVE and Δ NII Sensitivities

16.1 Introduction

One of the most complex challenges in **Asset & Liability Management (ALM)** is balancing the short-term profitability objective—captured by changes in **Net Interest Income (Δ NII)**—with the long-term **economic value sensitivity (Δ EVE)** of the balance sheet. Both indicators are key under the **IRRBB framework (EBA, 2022)**, yet they often move in opposite directions: actions that stabilize Δ EVE may reduce short-term earnings, and vice versa.

This case study analyzes the **duration gap management strategy** of a medium-sized universal bank (“**Bank D**”) over a one-year horizon, exploring the quantitative trade-offs and operational levers available to optimize interest rate risk exposure.

16.2 Bank D – Balance Sheet Overview (as of Q2 2023)

Balance Sheet Item	Amount (€bn)
Customer Loans	82
Fixed Income Portfolio (HTC + FVOCI)	18
Interbank Assets	7
Total Assets	107
Customer Deposits	71
Term Funding	17
Interbank Funding	6
Equity	13
Total Liabilities + Equity	107

Key Interest Rate Risk Metrics:

- Duration Gap: **+0.40 years**
- Δ EVE (–200 bps): **–€62m**

- SOT EVE: **-7.5%**
- Δ NII (-200 bps): **-€25m**
- SOT NII: **-4.9%**

16.3 Problem Statement

Bank D operates in an environment of **rising long-term rates** but **stabilizing short-term rates**, with the **2Y-10Y spread** moving from -40 bps to +30 bps.

The ALCO has observed the following issues:

1. The **EVE sensitivity** remains high due to a positive duration gap.
2. **Short-term profitability (Δ NII)** is under pressure from rising funding costs.
3. Liquidity ratios (LCR, NSFR) are comfortably above regulatory thresholds.

The bank faces a strategic decision:

- **Option 1: Reduce duration gap** through short-term hedging (flatten EVE exposure).
- **Option 2: Increase duration gap** to capture NII upside in a potential rate stabilization phase.

16.4 Scenario Setup and Assumptions

Market Context (2023–2024)

Variable	Q2 2023	Q3 2023	Q4 2023	Q1 2024
ECB Deposit Rate (%)	4.00	4.25	4.50	4.25
EUR Swap 2Y (%)	3.40	3.30	3.10	2.90
EUR Swap 10Y (%)	3.00	3.10	3.20	3.25
Inflation (YoY, %)	6.1	5.4	4.9	4.1
GDP Growth (QoQ, %)	0.2	0.1	0.0	-0.1

Curve Dynamics:

After a period of sharp tightening, short-term rates plateau, while long-term yields drift upward — producing a **mild steepening**.

16.5 Strategy 1: Reducing the Duration Gap (Defensive)

The ALCO considers shortening the duration gap from **+0.40y to +0.20y**, aiming to **reduce Δ EVE volatility** and align the balance sheet with the regulatory “Outlier Test” (SOT EVE < –5%).

Operational Levers:

- Enter **€4bn pay-fixed / receive-floating IRS** (5Y) to hedge asset duration.
- Increase **short-term floating-rate assets** through corporate lending repricing.
- Shift **€2bn of FVOCI securities** into **HTC** category to reduce mark-to-market volatility.

Resulting Indicators:

Indicator	Before Strategy	After Strategy 1
Duration Gap (years)	+0.40	+0.20
Δ EVE (–200 bps, €mn)	–62	–38
SOT EVE (%)	–7.5	–4.6
Δ NII (–200 bps, €mn)	–25	–35
SOT NII (%)	–4.9	–6.2
LCR (%)	138	140
NSFR (%)	119	121

Interpretation:

Duration reduction enhances capital protection but sacrifices NII resilience.

The portfolio becomes less sensitive to long-term rate moves but more exposed to margin compression if rates decline.

16.6 Strategy 2: Expanding the Duration Gap (Offensive)

Alternatively, the bank could extend duration from **+0.40y to +0.65y**, betting on the stabilization of rates and higher medium-term margins.

Operational Levers:

- Purchase **€3bn of 5Y BTPs or covered bonds** (FVOCI).
- Enter **€2bn receive-fixed / pay-floating IRS** (7Y) to extend duration synthetically.
- Lengthen **asset-liability repricing** by issuing **3Y fixed-rate term deposits** to lock in funding cost.

Resulting Indicators:

Indicator	Before Strategy	After Strategy 2
Duration Gap (years)	+0.40	+0.65
ΔEVE (–200 bps, €mn)	–62	–78
SOT EVE (%)	–7.5	–9.4
ΔNII (–200 bps, €mn)	–25	–12
SOT NII (%)	–4.9	–2.5
LCR (%)	138	135
NSFR (%)	119	117

Interpretation:

Expanding duration enhances profitability resilience (NII sensitivity halves) but increases capital volatility, potentially breaching the internal risk limit for ΔEVE .

A proper hedge accounting strategy (macro fair value hedge) would be required to neutralize the accounting P&L noise.

16.7 Quantitative Trade-Off Analysis

Scenario	ΔEVE / Tier 1 (%)	ΔNII / Tier 1 (%)	IRRBB Profile	ALM Strategy Classification
Baseline	–7.5	–4.9	Moderate Risk	Neutral
Strategy 1 (Defensive)	–4.6	–6.2	Low Capital Risk	Conservative
Strategy 2 (Offensive)	–9.4	–2.5	High Capital Risk	Aggressive

Risk–Return Curve Interpretation:

The three configurations can be plotted in the **ΔEVE – ΔNII plane** as a convex trade-off frontier. The optimal position depends on the bank's **risk appetite**, **macro outlook**, and **regulatory capital flexibility**.

16.8 Discussion

1. **Dynamic Rebalancing** – Duration management is not a one-off decision; it requires dynamic adjustment in response to yield curve movements and customer behavior.

2. **Curve Shape Awareness** – The steepening or flattening of the yield curve influences both NII and EVE asymmetrically.
3. **Integration with Hedge Accounting** – Macro fair value hedge (MFVH) programs allow banks to neutralize accounting volatility when extending duration.
4. **Governance and Risk Appetite** – ALCO decisions should consider the *total risk budget*, including credit spread risk and liquidity cost of hedges.
5. **Regulatory Context** – EBA (2022) Guidelines emphasize the need for coherent internal metrics linking ΔEVE and ΔNII to ICAAP and stress testing.

16.9 Conclusion

Duration gap management remains a **cornerstone of ALM discipline**.

While quantitative models can suggest an “optimal” point on the ΔEVE – ΔNII frontier, the real optimization occurs through **strategic dialogue** within ALCO—balancing earnings stability, capital preservation, and investor confidence.

As markets evolve, **integrated frameworks** combining behavioral modeling, macroeconomic forecasting, and hedge accounting optimization will increasingly define the competitive edge in modern ALM.

Chapter 17 – Case Study 5: Liquidity Buffer Optimization and the LCR/NSFR Trade-Off

17.1 Introduction

Liquidity management is one of the foundational pillars of **Asset & Liability Management (ALM)**, ensuring that a bank can meet its short-term obligations while maintaining balance sheet efficiency.

Under the **Basel III framework**, liquidity is measured through two key regulatory indicators:

- the **Liquidity Coverage Ratio (LCR)**, which assesses short-term resilience to stressed funding scenarios, and
- the **Net Stable Funding Ratio (NSFR)**, which measures structural funding stability over a one-year horizon.

Although both aim to enhance financial stability, their interaction often creates **strategic trade-offs**:

increasing the liquidity buffer typically improves LCR but worsens NSFR, as high-quality liquid assets (HQLA) consume stable funding.

Conversely, extending liabilities to improve NSFR may reduce short-term flexibility and margin efficiency.

This case study examines how a large universal bank (“**Bank E**”) can optimize its **liquidity buffer composition** and **funding profile** to balance profitability, regulatory compliance, and balance sheet resilience.

17.2 Bank E – Balance Sheet Overview (as of Q4 2023)

Balance Sheet Item	Amount (€bn)
Cash & Central Bank Reserves	45
Sovereign Bonds (HQLA Level 1)	65
Corporate Bonds (HQLA Level 2A/2B)	22
Loans to Customers	310
Other Assets	58
Total Assets	500
Customer Deposits	280
Wholesale Funding (<1y)	45
Term Funding (>1y)	90
Capital & Reserves	85
Total Liabilities + Equity	500

Key Liquidity Metrics:

- LCR: **126%**
- NSFR: **103%**
- Cost of Liquidity Buffer: **0.42%** (weighted average spread vs. OIS)

17.3 Problem Statement

Bank E has maintained an ample liquidity buffer in response to regulatory pressures post-2022, when rate volatility and geopolitical tensions raised liquidity risk premia. However, the **marginal return on excess liquidity has deteriorated**, leading to NII compression and higher cost of carry.

The **ALCO** now faces the strategic challenge of re-optimizing liquidity holdings while keeping both regulatory ratios comfortably above 100%, without compromising funding flexibility or reputational stability.

17.4 Scenario Setup and Market Context

Variable	Q4 2023	Q1 2024	Q2 2024 (Forecast)
ECB Deposit Rate (%)	4.50	4.25	4.00
3M Euribor (%)	4.20	4.05	3.85
10Y Swap Rate (%)	3.25	3.30	3.40
Euro Area Inflation (YoY, %)	4.9	4.1	3.7
10Y BTP Yield (%)	4.25	4.35	4.40
3M–10Y Spread (bps)	+105	+130	+155

Macro Environment:

A soft-landing scenario with disinflation and moderate GDP slowdown is expected. Yield curves are steepening slightly, improving the opportunity cost of holding long-duration HQLA but reducing the incentive to maintain very high overnight liquidity balances.

17.5 Strategy 1: Liquidity Buffer Compression (LCR Optimization)

Bank E evaluates a strategy aimed at **reducing excess HQLA** by €25bn, reallocating part of the portfolio into **short-term lending** and **secured interbank assets**.

This strategy enhances profitability but requires careful calibration to avoid breaching LCR thresholds.

Operational Levers:

- Redeem **€10bn** of Level 1 sovereign bonds.
- Reduce **€15bn** of central bank deposits.
- Reallocate €20bn into **short-dated interbank assets (AAA counterparties)**.
- Maintain the remaining liquidity buffer aligned with regulatory minimums.

Resulting Indicators:

Indicator	Before	After Strategy 1
LCR (%)	126	112
NSFR (%)	103	106
Cost of Liquidity Buffer (%)	0.42	0.36
Average Asset Yield (%)	2.10	2.35
ΔNII (12-month horizon, €mn)	—	+185

Interpretation:

Reducing the liquidity buffer boosts net interest income by reallocating idle cash into yield-bearing assets.

The NSFR benefits marginally due to better term structure matching.

However, the reduced LCR margin may increase vulnerability under stress scenarios and could attract supervisory scrutiny if risk appetite is not well documented.

17.6 Strategy 2: Structural Funding Extension (NSFR Optimization)

Alternatively, Bank E could focus on **lengthening funding maturities** to reinforce structural stability and improve NSFR at the expense of short-term returns.

Operational Levers:

- Issue **€10bn 3-year covered bonds**.
- Replace €8bn short-term wholesale funding with **retail term deposits** (2-3 years).
- Retain liquidity buffer unchanged to preserve LCR comfort zone.

Resulting Indicators:

Indicator	Before	After Strategy 2
LCR (%)	126	124
NSFR (%)	103	111
Average Funding Cost (%)	1.75	1.95
ΔNII (12-month horizon, €mn)	—	–95
Liquidity Coverage Volatility (bps)**	—	↓ 20%

Interpretation:

The bank gains structural funding resilience and reduces liquidity volatility, at the cost of lower NII due to higher long-term funding costs.

This trade-off is typical in a regulatory context emphasizing stability over profitability.

17.7 Comparative Analysis of Strategies

Metric	Baseline	Strategy 1: LCR Optimization	Strategy 2: NSFR Optimization
LCR (%)	126	112	124
NSFR (%)	103	106	111
ΔNII (€mn)	—	+185	–95
Cost of Liquidity Buffer (%)	0.42	0.36	0.42
Weighted Avg. Funding Cost (%)	1.75	1.70	1.95
Volatility of Liquidity Ratios	Base	↑ Moderate	↓ Low
Strategic Posture	Opportunistic	Conservative	Defensive

Interpretation:

- Strategy 1 aligns with a **proactive yield optimization** philosophy suitable for benign markets and strong liquidity positions.
- Strategy 2 reflects a **stability-oriented approach**, better suited for periods of macro uncertainty or tightening market liquidity.

17.8 Governance and Implementation Considerations

1. **ALCO Oversight** – Both strategies must be supported by liquidity stress testing under various funding and market shocks.
2. **ICAAP / ILAAP Integration** – EBA guidelines (2022) require linking liquidity metrics to capital planning and internal stress horizons.
3. **Behavioral Modelling** – Stability assumptions on retail deposits and roll-off rates on market funding are crucial to NSFR reliability.
4. **Market Timing** – The cost of term funding is highly cyclical; timing issuance windows can materially alter outcomes.

5. **Interaction with Interest Rate Risk** – Funding and liquidity adjustments modify the duration and repricing profile of the balance sheet, thus influencing ΔEVE and ΔNII indirectly.

17.9 Quantitative Sensitivity Simulation

Shock Scenario	ΔLCR (bps)	$\Delta NSFR$ (bps)	ΔNII (€/mn)	ΔEVE (€/mn)
+100 bps Parallel Up	+120	+30	+75	–45
–100 bps Parallel Down	–140	–35	–210	+30
30% Retail Outflow Shock	–1,500	–40	–60	—

Interpretation:

Under moderate rate shocks, LCR improves in rising rate environments due to higher reinvestment returns, but stress liquidity shocks (outflows) remain the dominant risk factor. Stress testing confirms the importance of maintaining a **minimum LCR >110%**, even when optimizing returns.

17.10 Conclusion

Liquidity management in modern ALM is a **balancing act between resilience and profitability**.

While the regulatory framework has significantly reduced systemic liquidity risks, it also forces institutions to manage an **intrinsic tension** between the LCR and NSFR.

Strategically, the **optimal liquidity configuration** lies not at the maximum of either ratio but along an **efficient frontier**—where incremental liquidity safety is offset by measurable economic cost.

The ability to dynamically manage this frontier through data-driven liquidity forecasting, behavioral analytics, and funding diversification is a defining feature of **best-in-class ALM practice**.

Chapter 18 – Case Study 6: Hedging Fixed-Rate Assets under the IFRS 9 Macro Fair Value Hedge Framework

18.1 Introduction

Interest rate volatility represents one of the most significant sources of valuation and earnings risk for banks. In particular, **fixed-rate assets**—such as long-term mortgages, leasing contracts, and medium-to-long-term corporate loans—expose banks to **fair value risk** when market rates fluctuate.

Under the **IFRS 9 accounting framework**, banks can apply **Macro Fair Value Hedge (MFVH)** accounting to mitigate such risks, aligning the measurement of hedging derivatives and hedged items to avoid artificial volatility in the income statement.

This chapter explores the **design, implementation, and implications** of a macro fair value hedge strategy applied to a portfolio of fixed-rate loans. The case study focuses on “**Bank F**,” a medium-sized European bank seeking to stabilize its **economic value of equity (EVE)** and its **net interest income (NII)** profile amid a scenario of declining interest rates.

18.2 Regulatory and Accounting Context

Under **IFRS 9**, a **fair value hedge** aims to offset changes in the fair value of an exposure attributable to a specific risk—in this case, interest rate risk.

For banks, applying the **Macro Fair Value Hedge (MFVH)** model enables:

- Consistent accounting treatment between hedged items (e.g., fixed-rate assets) and hedging instruments (e.g., interest rate swaps);
- Reduction of **P&L volatility** arising from fair value changes of derivatives;
- Alignment with internal ALM risk metrics such as **ΔEVE** and **ΔNII**.

EBA and ESMA guidelines (EBA/GL/2020/07; ESMA Public Statement 2021/353) emphasize the need for consistent risk identification, documentation of the hedge relationship, and robust testing of hedge effectiveness—both **ex ante** (prospective) and **ex post** (retrospective).

18.3 Bank F – Initial Balance Sheet and ALM Profile (Q3 2023)

Item	Amount (€bn)	Avg. Yield/Cost (%)	Duration (years)
Fixed-rate Mortgages	12.0	3.20	5.2
Corporate Loans (fixed)	4.5	3.45	4.8
Variable-rate Loans	6.0	4.10	0.5
Securities (HTC)	3.0	3.00	3.0
Other Assets	4.5	—	—
Total Assets	30.0	—	—
Demand Deposits	11.0	0.15	0.1
Term Deposits	5.0	1.50	2.0
Wholesale Funding	6.0	3.10	1.5
Equity	8.0	—	—
Total Liabilities + Equity	30.0	—	—

Risk Metrics (Pre-Hedging):

- Duration Gap: **+1.05 years**
- ΔEVE (–200 bps shock): **–€115 million**
- ΔNII (–200 bps shock): **–€38 million**
- SOT EVE: **–12.4%**
- SOT NII: **–6.8%**

18.4 Scenario and Market Context

Variable	Q3 2023	Q4 2023 (Forecast)	Q1 2024 (Forward)
ECB Deposit Rate (%)	4.25	4.00	3.50
2Y EUR Swap (%)	3.65	3.35	3.00
10Y EUR Swap (%)	3.05	2.70	2.50
Inflation (YoY, %)	5.2	4.3	3.8
2Y–10Y Slope (bps)	–60	–65	–50

Interpretation:

Markets anticipate a **progressive easing cycle** from the ECB starting in early 2024, with forward curves pricing roughly **75–100 bps of cuts** over the next 12 months.

This environment poses a **downside risk** for banks with positive duration gaps and significant fixed-rate exposures, as falling rates would increase the fair value of liabilities relative to assets, depressing EVE.

18.5 Strategic Objective

Bank F aims to **neutralize part of its positive duration exposure** through macro fair value hedging, mitigating sensitivity to falling rates while preserving flexibility for future portfolio repricing.

The target is to reduce ΔEVE sensitivity from –12.4% to approximately **–6.0%**, corresponding to a reduction of about €55 million in value volatility.

18.6 Hedge Design and Operational Setup

Hedged Item:

A macro portfolio of fixed-rate loans (mortgages + corporate) with nominal value **€16.5bn** and average modified duration **5 years**.

Hedging Instrument:

Receive-fixed/pay-floating **Interest Rate Swaps (IRS)** with average notional of **€8bn** and average maturity of **4.5 years**.

The swaps are designated in a **Macro Fair Value Hedge relationship** under IFRS 9.

Operational Steps:

1. Identification of the macro-hedged portfolio;

2. Definition of the hedging objective (“offset fair value exposure to interest rate risk”);
3. Hedge documentation and designation at inception;
4. Implementation of **prospective and retrospective effectiveness tests** (ΔPV correlation > 80%);
5. Accounting entries for both derivative and hedged item adjustments through P&L.

18.7 Quantitative Effects – Base and Hedged Positions

Metric	Pre-Hedge	Post-Hedge
Duration Gap (years)	+1.05	+0.45
ΔEVE (–200 bps, €mn)	–115	–54
ΔNII (–200 bps, €mn)	–38	–30
SOT EVE (%)	–12.4	–6.0
SOT NII (%)	–6.8	–5.1
Hedge Effectiveness (R^2)	—	0.89

Interpretation:

The macro hedge achieves approximately **50% risk reduction** in ΔEVE while maintaining NII resilience.

The slightly higher short-term NII volatility stems from the IRS floating leg repricing faster than loan yields but remains within acceptable tolerance limits.

18.8 Accounting Implications

Under IFRS 9:

- The **interest rate swaps** are measured at **fair value through P&L**.
- The **hedged fixed-rate loans** are adjusted for **changes in fair value** attributable to the hedged risk, recognized symmetrically in P&L.
- Any ineffectiveness (difference between derivative and hedged item movements) is recorded directly in the income statement.

In this case:

- **Effective hedge ratio:** 0.91
- **Cumulative fair value adjustment (hedged item):** +€52.5mn

- **Cumulative fair value change (IRS):** –€51.0mn
- **Ineffectiveness recognized:** –€1.5mn

This low level of ineffectiveness (<3%) confirms the hedge's accounting validity and regulatory acceptability.

18.9 Alternative Strategy: Cash Flow Hedge (CFH) Approach

Bank F also considered a **cash flow hedge** strategy to stabilize future interest margins rather than fair values.

However, the **long maturity** and **fixed-rate nature** of the loan book make CFH less suitable, as it would not address valuation volatility directly impacting CET1.

Moreover, the **MFVH model** allows dynamic rebalancing without de-designation, consistent with ECB and EBA supervisory expectations.

18.10 Interaction with Risk and Capital Metrics

Metric	Before Hedge	After Hedge	Impact
CET1 Ratio (%)	13.8	14.0	+0.2
RWAs (Liquidity and Market Risk, €bn)	172	175	+3
Volatility of OCI (%)	2.5	1.1	↓ 56%
LCR (%)	132	130	–2
NSFR (%)	105	106	+1

Comment:

The macro hedge marginally increases market RWAs due to the derivatives' exposure but improves **capital stability** by dampening valuation shocks.

The overall risk-adjusted return on capital (RAROC) improves by approximately **45 bps**.

18.11 Supervisory and Governance Aspects

EBA and ECB stress the importance of linking macro hedging frameworks with ALM governance and model risk control.

Key compliance actions include:

1. **Comprehensive documentation** of the hedged portfolio and risk management intent (IFRS 9.6.4.1–6.4.3).
2. **Regular effectiveness testing** (regression or sensitivity-based methods).
3. **Integration within ICAAP/ILAAP**, ensuring consistency between accounting hedges and economic ALM metrics.
4. **Clear governance** through ALCO oversight and internal audit review.

Recent EBA reports (EBA/REP/2022/15) highlight that successful macro hedging requires close alignment between **accounting, risk management, and treasury functions**.

18.12 Conclusion

This case demonstrates how **Macro Fair Value Hedge accounting** can effectively mitigate valuation volatility for fixed-rate assets in a falling rate environment, improving both **economic value stability** and **capital predictability**.

By combining robust hedge documentation, regular effectiveness testing, and clear governance, the bank achieves both **accounting and economic coherence**.

In a broader strategic context, **macro fair value hedging** represents a key tool in the **ALM optimization toolkit**, bridging regulatory expectations, accounting discipline, and risk-adjusted performance objectives.

Chapter 19 – Case Study 7: Behavioral Modelling of Non-Maturing Deposits (NMDs) and Interest Rate Sensitivity

19.1 Introduction

Among the most distinctive challenges in bank Asset & Liability Management (ALM) is the treatment of **Non-Maturing Deposits (NMDs)**—such as current accounts, savings accounts, and sight deposits—which lack contractual maturities yet exhibit **stable behavioral patterns**.

Unlike term deposits, NMDs do not have a defined repricing date. Their economic duration depends on depositor behavior, interest rate sensitivity, and competitive dynamics in retail funding markets.

Therefore, accurate modeling of NMDs is essential to correctly assess a bank's **interest rate risk in the banking book (IRRBB)** and ensure alignment with **regulatory (EBA/GL/2018/02)** and **accounting frameworks (IFRS 9)**.

This chapter presents a **quantitative and behavioral case study** on how a mid-sized retail bank—**Bank G**—models its NMD portfolio to estimate duration, elasticity to interest rates, and implications for both **ΔNII** and **ΔEVE** metrics.

19.2 Conceptual Framework

NMDs represent a **hybrid liability**:

- From a contractual perspective, they are repayable on demand.
- From a behavioral perspective, a portion of them behaves as **“core deposits”**, providing stable, long-term funding to the bank.

To capture this duality, the ALM function typically estimates:

1. **Core vs. Volatile share** of deposits;
2. **Effective duration** (behavioral maturity) of the stable component;
3. **Repricing elasticity** to market rate changes (pass-through rate).

According to the EBA and Basel guidelines (BCBS 368, 2016), modeling must ensure:

- Stability and statistical significance of behavioral estimates;
- Periodic back-testing and validation;
- Consistency with liquidity metrics (LCR, NSFR).

19.3 Bank G – Portfolio Overview (as of Q2 2024)

Deposit Type	Volume (€bn)	Average Rate (%)	Share of Total (%)	Core Share (%)	Behavioral Duration (years)
Retail Current Accounts	10.0	0.25	50	70	3.5
Retail Savings Accounts	6.0	0.40	30	60	2.8
Corporate Sight Accounts	3.0	0.10	15	40	1.5
Other Operational Accounts	1.0	0.15	5	50	2.0
Total NMDs	20.0	0.27 (avg)	100	63 (weighted)	2.9 (avg)

19.4 Behavioral Modelling Methodology

Bank G employs a **multi-factor econometric approach** integrating both **historical time-series** and **panel regression analysis** to estimate behavioral parameters:

$$r_{dep,t} = \alpha + \beta_1 \cdot r_{mkt,t} + \beta_2 \cdot \Delta r_{mkt,t} + \beta_3 \cdot \pi_t + \varepsilon_t$$

Where:

- $r_{dep,t}$: average deposit rate at time t ;
- $r_{mkt,t}$: market reference rate (e.g., 3M Euribor);
- π_t : inflation rate;
- β_1 : long-run pass-through coefficient;
- β_2 : short-run rate adjustment sensitivity.

Empirical results (2012–2024 sample):

- Long-term pass-through (β_1) $\approx 0.30 \rightarrow$ only 30% of market rate changes are transferred to customers;
- Short-term adjustment (β_2) $\approx 0.12 \rightarrow$ gradual repricing effect;
- Inflation correlation (β_3) $\approx 0.10 \rightarrow$ partial sensitivity to price dynamics.

19.5 Stability and Core Deposit Estimation

Using a **survival function** approach, the bank estimates the **average life (AL)** of deposits as:

$$AL = \int_0^{\infty} S(t) dt$$

where $S(t)$ represents the probability that a deposit remains outstanding at time t . The model results indicate that roughly **63% of total NMDs** behave as stable core deposits with an **average behavioral maturity of 3 years**.

Bank G validates these results via:

- **Cohort tracking** (withdrawal probabilities over time);
- **Stress-testing** under rate shocks (± 200 bps);
- **Benchmark comparison** with peer institutions and EBA calibration tables.

19.6 Scenario Analysis: Interest Rate Normalization

Macro assumptions (2024–2025 forecast):

Variable	Current	+200 bps Shock	–200 bps Shock
ECB Deposit Rate	4.25%	6.25%	2.25%
3M Euribor	4.10%	6.10%	2.10%
Inflation (YoY)	3.5%	4.0%	3.0%

Model outputs for NMD portfolio:

Metric	Base	+200 bps	–200 bps
Weighted Avg. Deposit Rate	0.27%	0.87%	0.09%
Core Share (%)	63	55	68
Duration (years)	2.9	2.1	3.6
Δ NII (€mn)	—	–62	+48
Δ EVE (€mn)	—	–145	+122

Interpretation:

- Rising rates reduce deposit duration and stability as clients migrate to higher-yield products, increasing funding cost volatility.

- Falling rates extend deposit duration and improve NII resilience, as pass-through to customers is partial.
- Asymmetry in Δ NII and Δ EVE reflects depositors' nonlinear behavioral response—a clear manifestation of **convexity in funding behavior**.

19.7 Integration with ALM and Risk Governance

EBA and Basel frameworks require that behavioral assumptions be:

- **Documented** and approved by the **ALCO**;
- **Validated annually** through statistical back-testing;
- **Reflected consistently** in ICAAP and ILAAP stress scenarios.

In Bank G, the **Treasury/ALM unit** collaborates with **Risk Management** to:

1. Incorporate NMD behavioral durations into **repricing gap and EVE models**;
2. Adjust **FTP (Funds Transfer Pricing)** to reflect the cost of core deposits;
3. Align **IRRBB reporting** (Δ EVE and Δ NII sensitivities) with the latest behavioral calibration.

19.8 Practical Implications for Hedge Strategy

Understanding NMD behavior directly influences hedging and structural balance sheet decisions:

Strategic Option	Objective	Potential Drawbacks
Enter IRS pay-fixed swaps	Reduce positive duration gap	Exposure to falling rates
Extend asset duration (e.g., longer-term securities)	Benefit from stable funding base	Higher market risk & capital absorption
Hedge volatility with swaptions	Protect against convexity in NMD behavior	Option cost & accounting complexity
Increase term funding share	Reduce behavioral uncertainty	Reduced flexibility, higher liquidity cost

By recognizing the **asymmetric convexity of NMD behavior**, ALM managers can construct **dynamic hedging programs** that combine derivatives and funding mix optimization.

19.9 Quantitative Summary

Indicator	Pre-Model	Post-Behavioral Modelling
Duration Gap (years)	+0.85	+0.55
ΔEVE (–200 bps) (€mn)	–210	–115
ΔNII (–200 bps) (€mn)	–68	–39
SOT EVE (%)	–10.8	–6.1
SOT NII (%)	–7.0	–4.3

Conclusion:

Behavioral modeling reduced estimated interest rate sensitivity by approximately **45% in EVE** and **40% in NII**, improving capital stability and risk-adjusted performance without altering the accounting classification of liabilities.

19.10 Concluding Remarks

This final case study demonstrates how behavioral analytics can profoundly reshape ALM decision-making.

NMDs, often seen as “free optionality” on the liability side, in reality embody **embedded convexity risk** and **nonlinear duration dynamics** that demand sophisticated modeling.

Accurate behavioral estimation not only refines IRRBB measures (ΔEVE , ΔNII) but also enhances strategic decisions on hedging, funding, and pricing.

Ultimately, incorporating behavioral models transforms ALM from a **static duration-matching exercise** into a **dynamic, data-driven discipline**—a defining trait of modern bank risk management.

Conclusion – The Evolution of ALM: From Static Measurement to Strategic Intelligence

1. Rethinking ALM as a Strategic Discipline

Over the past two decades, **Asset & Liability Management (ALM)** has evolved from a narrow interest rate risk measurement exercise into a **strategic management function** that bridges risk, finance, and business strategy.

The lessons drawn from the case studies in this book—spanning crises, unconventional monetary policy phases, and the behavioral shifts in funding—highlight a fundamental truth:

“The effectiveness of ALM lies not in predicting markets, but in preparing the balance sheet to remain resilient under any scenario.”

ALM is no longer confined to static duration gap reports or compliance-driven sensitivity analyses. Modern ALM demands **forward-looking, data-integrated, and multi-dimensional** decision frameworks capable of managing:

- Interest rate and convexity risk (ΔEVE , ΔNII , Vega);
- Structural liquidity and funding concentration;
- Capital efficiency and transfer pricing;
- Behavioral and optionality risk embedded in both assets and liabilities.

2. Lessons Learned from Practice

Across the simulations and empirical exercises presented, several **key operational insights** emerge:

1. **Balance sheet convexity matters as much as duration.** The experience of 2022–2023 demonstrated that large, unhedged convexity exposures—especially in fixed-rate assets—can severely amplify mark-to-market and capital impacts. Hedging must address not only the first-order (delta) but also second-order (gamma/vega) risks.
2. **Behavioral modelling is now indispensable.** Non-maturing deposits, prepayment options on loans, and callable securities create hidden asymmetries in sensitivity profiles. Quantitative behavioral models (e.g., NMD duration, prepayment elasticity) have become essential for realistic IRRBB measurement and FTP calibration.
3. **ALM and treasury must operate as a unified ecosystem.** The separation between “risk measurement” and “funding execution” is no longer viable. The most resilient banks—

across Europe and North America—are those that have integrated ALM insights directly into liquidity planning, hedging, and pricing strategy.

4. **Hedge accounting has become a performance tool, not just a compliance exercise.**

The evolution of IFRS 9 and the growing use of **macro fair value hedge accounting** have enabled banks to hedge structural interest rate exposures (e.g., fixed-rate loans, mortgage books) without introducing accounting volatility. Properly implemented, hedge accounting aligns **economic risk management** and **financial reporting**, allowing ALM to support P&L stability and capital optimization.

5. **Scenario discipline is the cornerstone of resilience.** The simulations across 2008, 2011, 2020, and 2023 prove that “perfect forecasts” do not exist. However, structured scenario planning—combining macro narratives with quantitative sensitivities—enables management to act faster and more coherently when shocks arise.

3. The New ALM Frontier: Digitalization, Regulation, and Intelligence

Looking ahead, three transformative forces are reshaping the ALM landscape:

(a) Basel IV and EBA-IRRBB Implementation

The next regulatory phase (Basel IV, CRR3/CRD6, EBA 2025 IRRBB technical standards) will embed **economic value sensitivity ($\Delta E V E$)** and **net interest income ($\Delta N I I$)** directly into supervisory capital expectations. Banks will need to demonstrate **model robustness**, **behavioral calibration**, and **hedging consistency** under both Pillar 2 and Pillar 3 disclosure frameworks. In this environment, ALM becomes not merely a control function, but a **strategic capital management lever**.

(b) Digitalization and AI Integration

Advanced data analytics, artificial intelligence, and machine learning are revolutionizing the ALM process:

- Predictive behavioral models for deposit stickiness and prepayment risk;
- Dynamic optimization algorithms for balance sheet structure;
- Real-time simulation engines linking treasury data to stress-testing frameworks.

The integration of AI allows ALM to move from static reports to **adaptive decision systems**, where scenarios are continuously recalibrated as market conditions evolve.

(c) ESG and Sustainable Balance Sheet Management

Sustainability considerations are entering the ALM perimeter. The transition toward **green funding instruments**, **ESG-linked loans**, and **climate stress testing** requires incorporating environmental risk factors into liquidity and interest rate risk models. The next generation of ALM will need to integrate **financial and non-financial risk**—a profound shift toward holistic balance sheet stewardship.

4. Toward an Integrated ALM Intelligence Framework

The future of ALM lies in **integration**—of data, of governance, and of strategic intent. An “ALM Intelligence Framework” can be envisioned around four interlocking pillars:

Pillar	Objective	Analytical Core	Outcome
Risk Analytics	Measure ΔEVE , ΔNII , Vega	Dynamic sensitivity models, behavioral calibration	Accurate, real-time risk quantification
Strategic Optimization	Allocate capital and duration	Efficient frontier between risk & return	Proactive balance sheet steering
Liquidity & Funding Integration	Link IRRBB and LCR/NSFR	Scenario-based liquidity modelling	Coherent short- and long-term funding
Governance & Reporting	Align ALCO, CFO, CRO decisions	Transparent, auditable risk communication	Unified strategic direction

By embedding this framework, banks can evolve ALM into a **strategic command center** for the entire balance sheet — one that synthesizes financial stability, profitability, and regulatory compliance.

5. Final Message

The journey of ALM—from duration matching in the 1980s, through VaR-based frameworks post-2008, to behavioral and AI-enhanced modelling today—reflects the evolution of banking itself.

In an environment characterized by rapid rate cycles, digital transformation, and regulatory tightening, **strategic ALM** becomes a source of **competitive advantage** rather than a mere compliance necessity.

The central message of this book is thus both simple and profound:

“Banks that treat ALM as an intelligence function—integrating risk, capital, liquidity, and behavior—will not only survive future rate shocks but will thrive on them.”

ABOUT THE AUTHOR

Alessio Gioia, born in Suzzara (Mantova) in 1971, married, two daughters. Graduated from the Catholic University of the Sacred Heart in Milan in 1996 in Banking, Financial and Insurance Sciences with a thesis on Asset Backed securitization. Entering, immediately after graduation, in the world of operational finance as a currency market trader with a broker, he had the opportunity to deepen the logic of approach to proprietary trading and portfolio management. Later he gained experience as a “prop trader and portfolio manager” in banks and asset management companies. In the recent past he worked as Head of ALM and Banking Book and investment manager for one of the most important Italian banks. He is currently General Bursar of the diocese of Mantua.